

Supplementary Material: Self-Play Can Mask Frontier LLM Risk Responses in AI Development Races

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Abstract

This supplement supports Self-Play Can Mask Frontier LLM Risk Responses in AI Development Races. It records the quality-control screen, the theory comparison, the scripted-rival evidence, and the human diversity analysis, together with prompt and game robustness checks. These materials use the same routes, task and evidence boundaries as the main paper and do not extend its claims beyond the idealised race.

CCS Concepts

• **Computing methodologies** → **Multi-agent systems**.

Keywords

Large language models, AI development races, multi-agent systems, repeated games, AI safety, opponent-conditioned behaviour

ACM Reference Format:

Anonymous Author(s). 2027. Supplementary Material: Self-Play Can Mask Frontier LLM Risk Responses in AI Development Races. In *Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027)*, Hanoi, Vietnam, 3–7 May 2027, IFAAMAS, 22 pages.

Supplementary roadmap

The supplement is organised as a short evidence package rather than a lab notebook. The matrix below makes the endpoint roster and evidence status explicit before the detailed checks begin.

For the headline claims, read the screen, scripted-opponent, EGT-fit and human rows first. Open-weight, pilot-roster, persona, embedding and group-size analyses are exploratory extensions and are not pooled with the eligible endpoint claims. The scripted-opponent block is physically placed after the robustness checks so its complete evidence follows the main paper’s rival result directly.

Four terms are used with fixed meanings throughout. An *endpoint* is the exact hosted model version and provider configuration tested, for example `google/gemini-3-flash-preview`. An eligible endpoint carries a passing eligibility outcome from the probe battery below. A *checkpoint* is used only for a dated model version when the distinction matters. *Model* is used generically for a language-model system. The screen outcome is either *screen-passing* (eligible) or *screen-failing* (excluded); it is not a capability score.

CONTENTS

Abstract

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Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027), M. Baldoni, F. Fang, W. Yeoh, N. Yorke-Smith (eds.), 3–7 May 2027, Hanoi, Vietnam. © 2027 International Foundation for Autonomous Agents and Multiagent Systems (www.ifaamas.org).

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Supplementary evidence

This supplement collects the material referenced from the main paper as supporting rather than extending the main claims: endpoint eligibility and provenance, risk-response and theory robustness, the crossed presentation check, the full scripted-opponent table, the human-benchmark replication and trajectory-diversity statistics, and the matched group-size extension. The historical pilot rosters, classifier, predictive analysis, player-level rate distributions and embedding diagnostics are included only to support the main paper’s scoped exploratory pointers. The separate open-weight battery, persona diagnostics, older pilot baselines, archetype projections, and joint-dynamics/horizon extensions remain in the repository as extended artifacts rather than part of this evidence package.

S1 Endpoint eligibility and provenance

S1.1 The quality-control screen: what it asked, and which routes passed

This subsection records the screen as an entry condition and nothing more: what it asked, when its rule was fixed, which routes

Table 1: Coverage matrix for the supplementary evidence. Endpoint counts are shown for the analysis that actually supports each claim; no row pools the nine-screen roster with the separate open-weight or pilot batteries.

Analysis	Endpoints	Evidence status	Main claim?
Self-play risk response	9 screened / 5 eligible	Confirmatory	Yes
Scripted opponent policy	5/5 eligible	Confirmatory	Yes
EGT direct comparison	2 prespecified	Confirmatory	Yes, direct
EGT secondary fit	5/5 eligible	Secondary reanalysis	Yes, robustness
Presentation robustness	2 prespecified	Confirmatory, scope-limited	Limited
Human trajectory diversity	9 screened / 5 eligible headline	Confirmatory baseline	Yes
Matched group size	1 eligible endpoint	Exploratory	No

cleared it, and the three facts about it that bound what a verdict may be used for. It is deliberately short. Nothing in this supplement ranks routes by their probe scores or treats the verdict as a measurement of capability.

The screen is a frozen bank of 20 probes over six domains, asked three times each, giving 60 scored answers per route and 540 in all. The six domains are recalling the rules, reading the stage-payoff matrix, rebuilding the current state from a short history, taking one state transition, scoring a finished race, and computing a closed-form expected payoff. A route is admitted only if it clears all three of the conditions recorded in the caption of Table 2: 80% of all its answers, 75% within state reconstruction and 75% within terminal scoring. Expected payoff is recorded and never blocks admission, because the game asks an agent to choose an action rather than to solve for an expected value.

Two properties of the rule are what let it serve as an entry condition. The bank, the three conditions and their values were fixed and recorded before the campaign ran, in the written collection protocol. And the screen scores probe answers against ground truth without ever reading a race, so no gameplay result can reach a verdict, whatever order things were run in. We do not call the screen pre-registered, because three of the nine routes received their first verdict after their own gameplay had completed; what is checkable, and what the defence rests on, is the fixed bank, the fixed rule and the fact that the instrument cannot see the outcome it selects on.

Three facts make the nine rows of Table 2 comparable with one another. All nine ran under one protocol identifier. All nine answered the same probe bank, whose SHA-256 hash begins 2953fb47, and read the same rules context, whose hash begins b80981a5; the analysis script fails closed and refuses to tabulate a route whose hashes differ. Eight routes ran at benchmark task version 7 and one, google/gemini-3.5-flash-lite, at version 8, whose outgoing request omits the reasoning-budget argument because that provider returns an invalid-argument error when the argument is present at all. That is a difference in the transport contract rather than evidence about the model, it is recorded in that route’s own manifest, and the amendment introducing per-route contracts is dated in the collection protocol. For every other route the outgoing request is byte-identical to the earlier runs. One failed run is retained rather than deleted: the same route at task version 7 ended in exhausted transport retries and produced no scoreable answer, so it appears nowhere in the table and is kept as a transport failure beside the campaign.

Two admitted routes pass by nothing at all. State reconstruction and terminal scoring are each scored over fifteen answers, so a 75% threshold asks for 11.25 correct and the lowest score that can clear it is 12 of 15, or 80.0%. GPT-5.5 clears state reconstruction at exactly that score and Claude Sonnet 5 clears terminal scoring at exactly that score, so both are admitted with a margin of zero. That is one reason a verdict is used here as a declared entry condition and never as a trait estimate. No analysis in this paper interpolates between probe scores, ranks admitted routes by probe accuracy, or reads the admitted against refused split as a measured contrast.

The split follows the tier word in the route name. All four refused routes carry mini, nano or lite as a whole word of their name, and none of the five admitted routes does. On this roster, then, the screen draws no line that the route name does not already draw. Whether a screen of this kind measures comprehension is outside this paper’s scope. This supplement uses the screen only as the entry condition it is.

Nothing was discarded. All nine routes played the same confirmatory baseline with the same race counts, and every refused route’s behaviour is reported in this supplement beside the five admitted ones. Four further advertised routes never produced a scoreable answer at all, because transport failed before any probe was sampled; they were retained as failure records and never replaced by a substitute endpoint.

S2 Risk-response robustness

S2.1 Earlier two-route confirmatory baseline

The new neutral gameplay baseline used protocol v3, the same mechanism and prompt for the two routes used in the direct EGT comparison, ten repetitions per risk level, and 30 races per route. Each route produced 558 dependent within-race decisions with zero parse failures. The entries in Table 3 are decision-weighted UNSAFE rates; they are not pooled across routes and do not estimate an evolutionary equilibrium.

Race-weighted sensitivity. The headline entries above weight each decision equally, matching the run summaries. Because both seats in a race share the same horizon draw, we also report the race as the independent unit. Table 4 repeats the risk-response contrast for the five admitted routes using equal weight per race and a percentile bootstrap over race clusters. The point estimates preserve the same route ordering and the intervals are a sensitivity check on weighting, not an additional model run.

Table 2: The nine screened routes on the three conditions that decide admission, and which condition refused each route that did not enter. A route is admitted only when all three clear: overall accuracy at least 80%, state reconstruction at least 75% and terminal scoring at least 75%. Every cell is the percentage of that domain’s probes answered correctly. Overall accuracy rests on all 60 scored answers per route, state reconstruction and terminal scoring on 15 each, so those two move in steps of one fifteenth, about 6.7 points. Expected payoff is recorded as a diagnostic and gates nothing; no route reaches 75% on it. Rule recall, payoff reading and state transition are not admission conditions and are not shown.

Exact route	Overall accuracy	State recon.	Terminal scoring	Verdict
google/gemini-3-flash-preview	93.3	93.3	100.0	admitted
anthropic/claude-opus-5@default	91.7	93.3	100.0	admitted
openai/gpt-5.4-2026-03-05	90.0	100.0	100.0	admitted
openai/gpt-5.5-2026-04-23	90.0	80.0	100.0	admitted
anthropic/claude-sonnet-5@default	85.0	100.0	80.0	admitted
google/gemini-3.1-flash-lite-preview	80.0	73.3	80.0	refused, state recon.
openai/gpt-5.4-mini-2026-03-17	75.0	66.7	86.7	refused, two conditions
google/gemini-3.5-flash-lite	65.0	60.0	60.0	refused, all three
openai/gpt-5.4-nano-2026-03-17	51.7	20.0	66.7	refused, all three

Table 3: Two-route confirmatory block used for the direct EGT comparison. Each route has 30 races, 558 dependent within-race decisions, and zero parse failures.

Exact route	$r = .1$	$r = .6$	$r = .9$	Races
google/gemini-3-flash-preview	98.9%	73.1%	60.2%	30
anthropic/claude-sonnet-5@default	89.2%	48.4%	32.3%	30

Table 4: Risk-response sensitivity to the weighting unit. The contrast is the decision-weighted or race-weighted UNSAFE rate at maximum private risk 0.1 minus the rate at 0.9. Race-weighted brackets are 95% percentile bootstrap intervals over ten race clusters per risk cell.

Admitted route	Decision-weighted	Race-weighted [95% RI]
Gemini 3 Flash	38.7	39.1 [35.8, 42.2]
Claude Opus 5	100.0	100.0 [100.0, 100.0]
GPT-5.4	40.3	40.7 [34.1, 47.4]
GPT-5.5	38.2	39.7 [33.1, 46.9]
Claude Sonnet 5	57.0	55.0 [50.4, 59.8]

The Gemini manifest records that the hosted SDK stripped the requested sampling seed. The Claude manifest records that the seed was forwarded but provider-side application was not confirmed. Temperature was requested as 0.7 for both routes but was not forwarded by this SDK path. These facts bound the result to the exact hosted route and software contract.

To make the comparison boundary explicit, Figure 1 separates the risk response, the route-specific departure from the strong-selection reconstruction, the nearest-rule diagnostic, and its mismatch rate. This is an analysis artifact rather than an equilibrium test: the frontier routes generate prompted self-play trajectories, whereas EGT samples stationary population composition.

Which strategy takes that population is a one-line result, now shown in Figure 2. Read off the archived EGTtools fixation probabilities, the strategy that dominates the independently simulated finite-mutation chain ($\beta = 2$, $\mu = 0.02$) switches with risk: Always UNSAFE at $p_r^{\max} = 0.1$ with 91.9% of the chain, the conditional rule that opens UNSAFE at 0.6 with 96.3%, and the conditional rule that opens SAFE at 0.9 with 93.9%. Exactly nine of the twelve off-diagonal

invasions per risk level clear neutral drift $1/Z = 0.01$, and the smallest fixation probability among those nine is 0.997, so the ordering is not marginal. Two boundaries travel with that sentence. The fixation probabilities are evaluated in the EGTtools small-mutation limit while the shares come from the finite-mutation chains, which are two different simulation layers; and no language-model route plays this population process, so none of it is a latent-strategy statement about any endpoint.

S2.2 The shape of a policy behind a rate

Every gameplay number in the neutral-baseline analysis is a rate: the share of decisions that came out UNSAFE. A rate is silent about the thing an iterated game is for. Two routes can sit at the same rate while one is reading the board and the other is reciting a constant, and no table in this supplement separates those two cases. Figure 3 is the object that does, and that is what it adds to the neutral baseline: not another number for how much UNSAFE play there was, but the policy that produced it.

The read-out is a memory-one action-response summary used in the evolutionary game theory literature: the probability of playing UNSAFE given the pair of moves that preceded it, the route’s own move first and its rival’s second. The choice is not an analytical convenience. The prompt exposes the immediately preceding round, but it also exposes current progress, accumulated risk and the round index. The four conditional probabilities therefore marginalise those other state variables; they summarise action-response regularity rather than identify a complete stationary policy. The summary is computed over the same nine-route neutral baseline the rest of this paper reports, 270 races and 5,022 decisions, of which the 4,482 that have a predecessor round enter the calculation; round one is excluded rather than folded in, because an opening move is a different object from a reply.

The two lagged associations beside it are Mantel-Haenszel differences, stratified on the risk level and on the other lagged move. Both stratifications are load-bearing. Risk has to be held fixed because the three risk cells have very different base rates, so pooling them would let a route that simply plays more UNSAFE at low risk look responsive. The other lagged move has to be held fixed because these are self-play races: both companies are the same route, so an

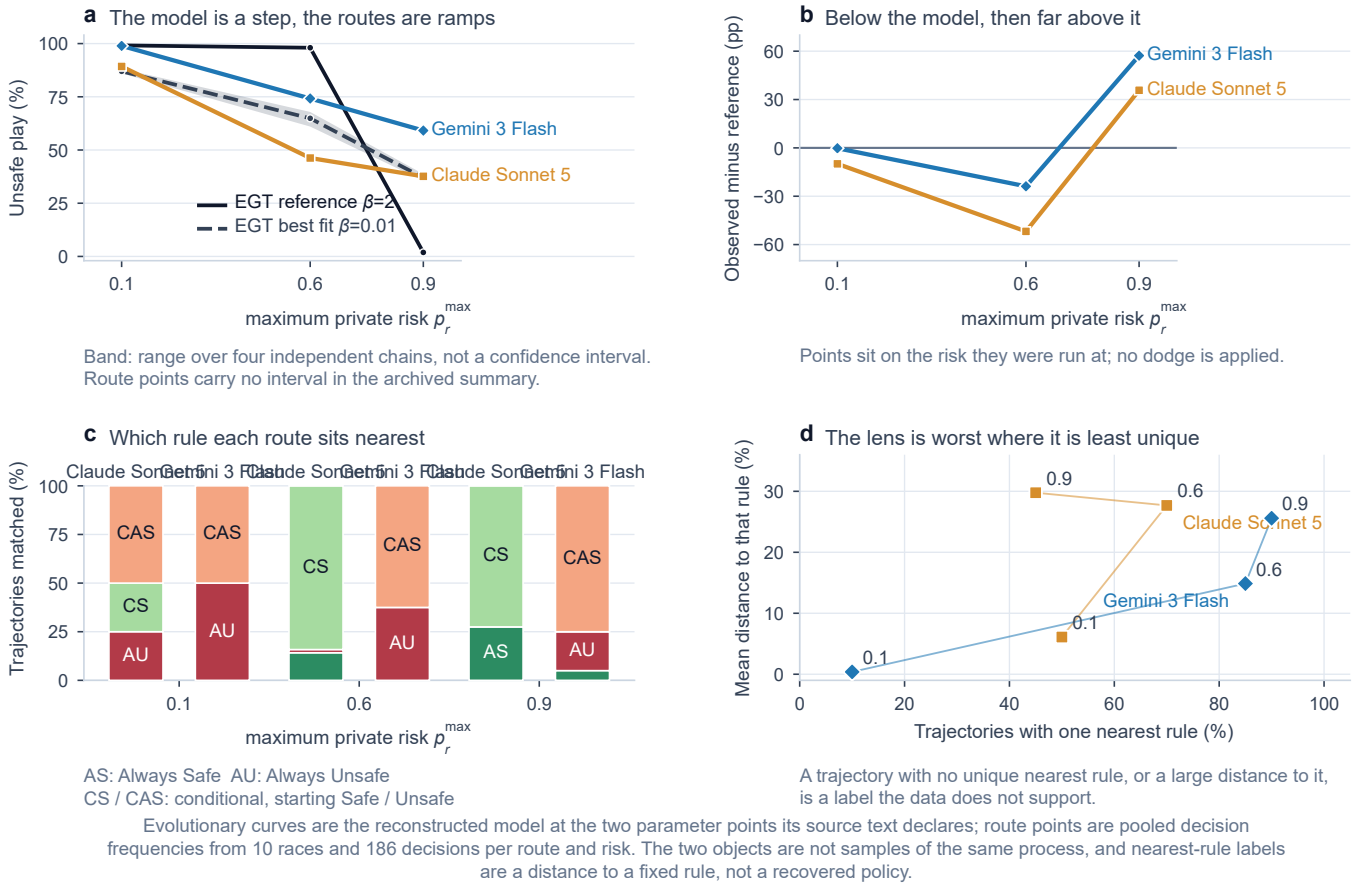


Figure 1: The reduced evolutionary model against the two routes that were run through the comparison protocol, anthropic/claude-sonnet-5@default and google/gemini-3-flash-preview. These are two of the five routes the quality-control screen admits, not the admitted set. (a) At its reference parameter point the model is a step in risk, holding UNSAFE play near the ceiling through $p_r^{\max} = 0.6$ and dropping it to 1.9% at 0.9; both routes are ramps and both stay far above the model at 0.9. The band on each model curve is the range across four independent seeded chains, which is a convergence diagnostic and not a confidence interval; the route points carry no interval because the archived summary holds none. (b) The same contrast signed against the reference, with each point on the risk it was run at. (c) Which of the four rules each route’s trajectories sit nearest, one stack per route. (d) Why (c) is a lens and never a finding: the labels are least unique exactly where the distance to the nearest rule is largest.

unstratified reading of what a route does after its rival went UNSAFE would largely measure the route’s own phase. Intervals resample races within their risk cell, the same clustering every other interval in this paper uses.

Reading the two panels together separates three kinds of route that a rate column would print side by side. Six of the nine answer their rival’s last move by moving toward it, and every route whose terms are defined at all has a negative own term, so the pattern is not momentum in either direction but matching the rival while correcting itself. On GPT-5.4 mini that own term is the one whose interval spans zero, -4.8 pp $[-12.7, +3.4]$, so seven of the eight are estimated away from zero and the eighth is not. Two routes, GPT-5.4 mini and GPT-5.4 nano, carry a rival term pointing the other way; they are also the two whose UNSAFE rate barely moves with the stated risk. Claude Opus 5 is the third kind: a step whose

two seats never diverged in 279 rounds, for which the question the fingerprint asks does not arise.

What it does not show. Responding to a rival is not understanding a rival, and nothing here licenses a claim about strategy, intent or reasoning: these are conditional frequencies in one game at one prompt version, on nine commercial endpoints that are not a sample from a population of models. The own term in particular is not a preference. After its own UNSAFE move a route is also carrying more accumulated private risk, which is a real feature of the game rather than a confound, but it means the term measures a response to the state and not to the move alone.

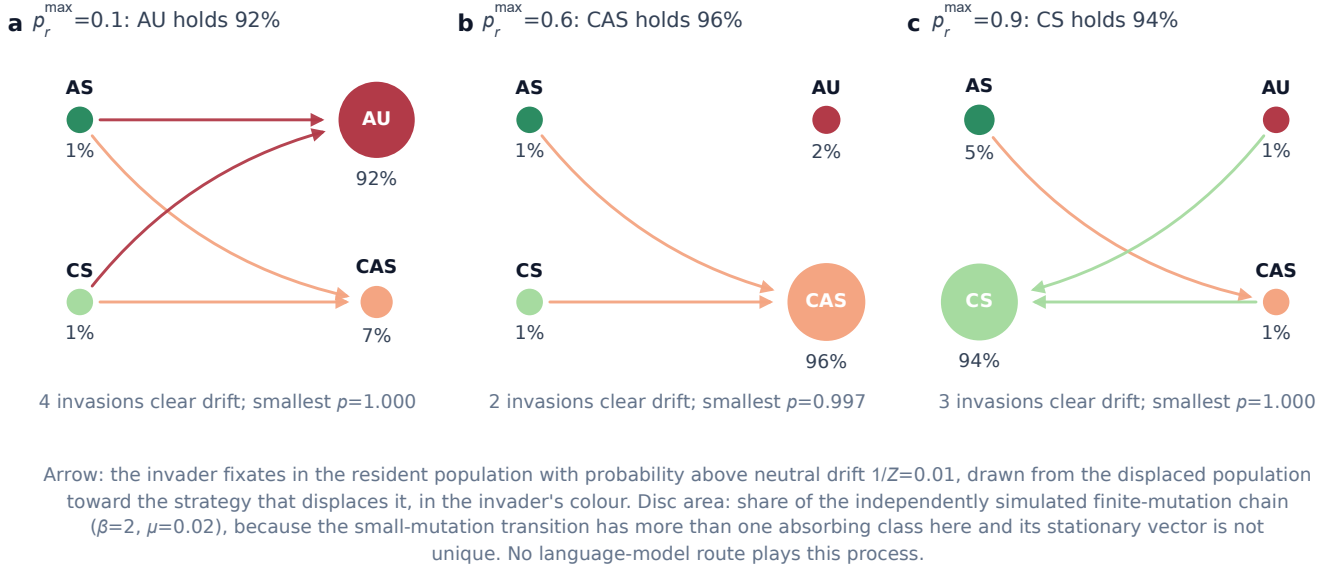


Figure 2: Risk-conditioned invasion topology in the reconstructed evolutionary game. Each panel is a standard invasion graph at one maximum private-risk level. The four nodes are Always Safe (AS), Always Unsafe (AU), Conditionally Safe (CS) and Conditionally Antisocial Safe (CAS); node area shows the share of the independently simulated finite-mutation population. An arrow points from the resident strategy to an invader whose EGTtools fixation probability exceeds neutral drift $1/Z = 0.01$. The graph shows which strategy can replace which resident, while node areas show the finite-mutation population composition.

S2.3 How much of a route's number is the run?

Every route-level rate in this paper comes from one thirty-race run, so a reader is entitled to ask how much of a rate is the route and how much is the particular run. One route answers that question directly. Sampling on this route is not reproducible. On the three Google routes the SDK strips the seed request before it reaches the provider, and every run manifest for those routes records that. On the OpenAI and Anthropic routes the SDK forwarded the seed, and no provider confirmed applying it. Two runs of the same Google cell are therefore genuinely independent draws from the route, not a rerun of one draw.

google/gemini-3-flash-preview was collected twice under the same frozen baseline protocol. The first administration is the run reported elsewhere in this paper; the second is an independently sampled repeat that was not planned as confirmatory evidence. Protocol, prompt hash, mechanism, seed structure and the full set of thirty (game seed, risk) blocks are identical across the two administrations, and the released analysis code refuses the comparison unless they are. The repeat is analysed separately and is never pooled with the primary campaign.

Agreement is close. UNSAFE rates move from 98.9% to 100.0% at $p_r^{\max} = 0.1$, from 73.1% to 74.2% at 0.6, and from 60.2% to 59.7% at 0.9: the largest per-risk difference over 186 decisions is 1.1 percentage points, and the risk response, the UNSAFE rate at risk 0.1 minus the rate at 0.9, moves from 38.7 to 40.3 percentage points. That is small beside the spread in risk response across the nine routes of the neutral baseline, which runs from 7.0 points on GPT-5.4 mini to 100.0 points on Claude Opus 5, a spread of 93.0 points. That between-route spread is the comparison this check exists to protect.

It is one route and one repeat, so it bounds run-to-run variation for this route at this sample size and licenses nothing for the routes that were collected once. The repeat is stored apart from the main campaign records, and the analysis fails if an already represented route could replace the primary run.

S3 Evolutionary benchmark sensitivity

S3.1 Selection-strength sweep of the evolutionary benchmark

The comparison in the direct-baseline analysis uses one setting of the reconstructed evolutionary model. A reviewer may reasonably ask whether the gap between that model and the audited routes is a property of the model or an artifact of that one setting. This subsection answers that question by sweeping the setting.

Two numbers govern the model. *Selection strength* β says how strongly a simulated agent prefers to copy a better-performing neighbour: at β near zero copying is almost random, and at large β it is almost deterministic. The *mutation rate* μ is the chance, per update, that the sampled individual abandons its strategy and switches uniformly to one of the other three. The sweep covers ten values of selection strength, $\beta \in \{0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1, 2, 3, 10\}$, crossed with three rules for setting the mutation rate: $\mu = \beta/Z$, the main-text convention, which gives $\mu = 0.02$ at $\beta = 2$; $\mu = 1/Z = 0.01$; and a fixed $\mu = 0.05$, the convention under which the source study reports its best fit. With $Z = 100$ individuals, four strategies, and the mechanism's three maximum-private-risk levels (0.1, 0.6, 0.9), the sweep contains 90 cells built from 29 distinct parameter pairs, since some pairs coincide across rules and were simulated once. Each cell is estimated from four independent chains of 400,000

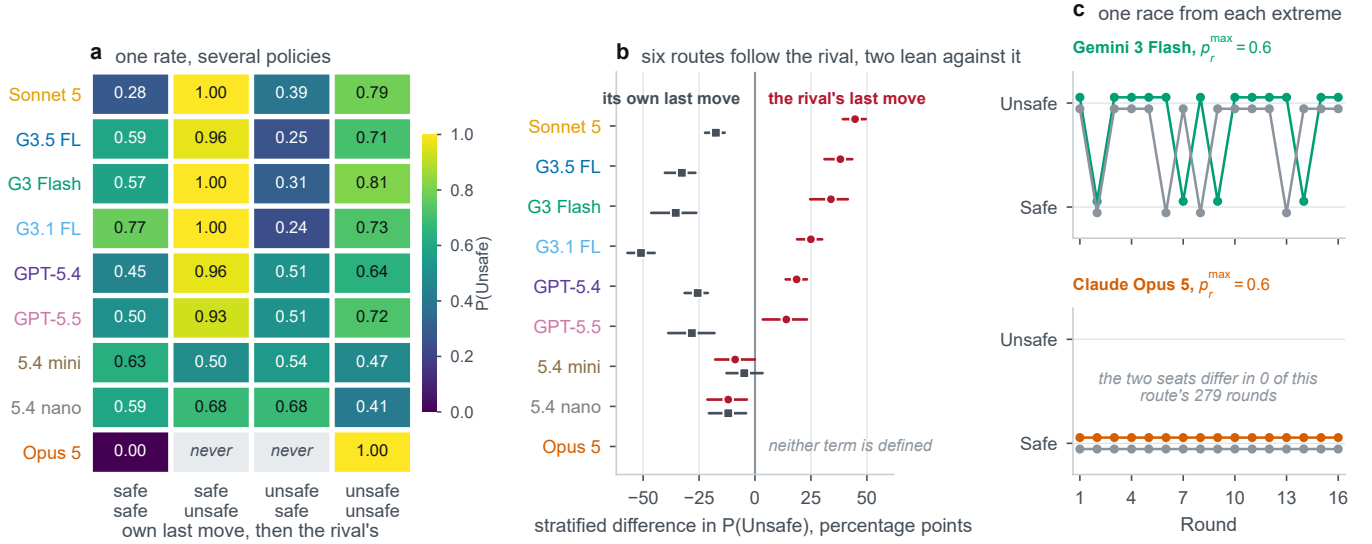


Figure 3: The memory-one policy behind each route’s UNSAFE rate, over the nine-route neutral baseline. (a) One row per route and one column per preceding pair of moves, the route’s own first and its rival’s second, with the conditional probability printed in every tile; row labels carry the route colours used throughout the paper, so the panel needs no legend. Two of Claude Opus 5’s tiles are marked absent rather than left blank: its two seats never once played differently, so the mixed contexts did not arise and there is nothing there to measure. (b) The two lagged associations on one scale, Mantel-Haenszel differences in $P(\text{UNSAFE})$ stratified on the risk level and on the other lagged move: how far UNSAFE play differs after the rival’s UNSAFE move, and how far it differs after the route’s own, with race-clustered 95% intervals and the routes in the same order as panel a. (c) Two races at the same risk level under the same protocol, one from a route that answers its rival and one from the route that does not move at all, with the two seats drawn separately. Every quantity here is an association within a stratum and not a causal effect. Nothing is randomised: these are conditional frequencies from nine commercial endpoints in one game at one prompt version, self-play throughout, and the own term in particular also carries the accumulated private risk that an earlier UNSAFE move leaves behind, so it is a response to the state and not to the move alone.

steps with a burn-in of 100,000, thinned by 100, giving 4,000 retained samples per chain.

The reference setting reproduces. At $\beta = 2$ with $\mu = \beta/Z = 0.02$, the sweep returns a decision-weighted UNSAFE share of 99.2%, 98.0%, and 1.9% at maximum risks 0.1, 0.6, and 0.9: the same three-point profile the main paper attributes to the published reference, matched to within 0.0002, 0.0001, and 0.0001 in share, far inside the 0.02 tolerance the check declares in advance. The sweep uses a deliberately disjoint seed scheme from the reproduction script, so this agreement is an independent-seed check rather than a rerun of the same random numbers.

The qualitative gap does not close anywhere on the grid at the reference selection strength, but it does close at very weak selection. Of the 90 cells, four reproduce the qualitative shape the admitted endpoints show, namely near-total UNSAFE play at risk 0.1, a middling rate at 0.6, and a lower but clearly non-zero rate at 0.9, declining at every step. Three of those four also pass the mixing diagnostic. All of them sit at weak selection: $\beta = 0.01$ and $\beta = 0.03$ under $\mu = 0.05$, and $\beta = 0.003$ and $\beta = 0.01$ under $\mu = 1/Z$. Ten of the 90 cells are flagged as mixing-suspect, meaning their four independent chains disagreed by more than 0.05 on the aggregate UNSAFE share, so their chain mean must not be read as a

stationary distribution; almost all of them are the very small mutation rates that $\mu = \beta/Z$ produces at small β , which make the four uniform-strategy states nearly absorbing.

The best-fitting cell is the same for both audited routes, and it is not the reference setting. Fitting is by root-mean-squared deviation, in percentage points, between the model’s three-point UNSAFE profile and the route’s observed decision-weighted UNSAFE rates at the same three risk levels; the largest single-condition deviation is reported beside it so that one badly missed condition stays visible. For both routes the minimum-error cell is $\beta = 0.01$ with fixed $\mu = 0.05$, whose profile is 87.3%, 63.9%, and 38.0%. Against anthropic/claude-sonnet-5@default, observed at 89.2%, 48.4%, and 32.3%, that cell has an error of 10.28 percentage points and a largest single-condition miss of 17.70 points, at risk 0.6. Against google/gemini-3-flash-preview, observed at 98.9%, 73.1%, and 60.2%, the error is 15.13 points and the largest miss 21.14 points, at risk 0.9. Even the best cell on the grid is therefore more than ten percentage points away from either route.

All-five admitted-route extension. The preceding block is the earlier two-route confirmatory comparison. To check whether its qualitative reading depends on that restricted roster, we fit the same existing 90-cell sensitivity table to the canonical neutral-baseline profiles of all five admitted routes. This is analysis only: no LLM

Table 5: Analysis-only EGT fit across all five admitted routes. “Reference” is the $\beta = 2$, $\mu = \beta/Z$ cell; “weak” is the best well-mixed cell with $\beta \leq 0.03$. Values are RMSE in percentage points.

Route	Observed shape	Reference	Best weak
Gemini 3 Flash	graded decline	36.6	15.4
GPT-5.4	graded decline	32.9	5.1
GPT-5.5	graded decline	34.9	11.3
Claude Sonnet 5	graded decline	34.1	9.6
Claude Opus 5	near-step switch	56.6	43.6

requests or evolutionary chains were added. Distances are RMSEs in percentage points over the three matched risk levels, and the primary cells below are required to be well mixed. The best weak-selection cell is the same in every route, $\beta = 0.01$ with fixed $\mu = 0.05$; the four graded routes are substantially closer to it than to the reference $\beta = 2$ (Table 5), while Claude Opus 5 remains a qualitative near-step exception. The restricted two-route block and this extension use distinct observed profiles, so the RMSEs for their two overlapping routes are not pooled and need not match.

One boundary is worth restating, because the reproduction lane states it itself. This is a repo-native reconstruction of the disclosed reduced evolutionary model, not a bitwise reproduction of the original authors’ code: that code, the generated payoff matrices, the payoff Monte Carlo seeds, and the exact library revision are not public in the preprint version we worked from, and the compiled comparison class could not be executed on this interpreter because no matching wheel exists for it, so the sweep ran the pinned pure-Python implementation instead. The quantities above are therefore seeded-chain estimates reported with their between-chain spread, not exact stationary distributions, and the route rates beside them are descriptive prompted self-play endpoints rather than draws from a population process.

Reading the model backwards. The sweep above asks what the model predicts. The complementary question is what an observed rate would tell us about the model’s own risk parameter, and Figure 4 asks it. The answer bounds how finely this benchmark can be read at all. Because the only place the model bends is its cliff, asking which risk would make it emit an observed rate sends nearly every answer to the same place: the three configured levels span 0.80 of the risk axis, and the 24 invertible route-by-risk cells come back inside a band 0.12 wide, a compression of about sevenfold. What is lost is resolution and not ranking, and the distinction matters. All eight invertible routes return their three implied risks in the configured order, and seven of the eight cells run at 0.1 land strictly left of every cell run at 0.9, the two extreme rows sharing only 0.0006 of the axis. But adjacent rows overlap, the 0.1 and 0.6 rows on 0.0147 and the 0.6 and 0.9 rows on 0.0129, so a rate read back through this model fixes an implied risk only to about a tenth of the axis it was configured on. The inversion runs at $\beta = 0.01$, the weak-selection point, which is the setting that gives the model its best chance of spreading the cells apart; at $\beta = 2$ the inverse is defined only inside a 0.005-wide window, and inverting there would manufacture the compression. Three limits travel with the figure. It is a statement about the model, not an estimate of any route’s perceived risk: it says only what the model would have to be handed. A compressed

Table 6: Fully crossed narrative-skin by symbol-mapping rerun, first of the two routes that completed it, google/gemini-3-flash-preview; the second route, anthropic/claude-sonnet-5@default, completed the same twelve cells, and its cell rates are released with the analysis code. “Mapping” names which response code stands for SAFE: under P_SAFE_Q_UNSAFE the code P means SAFE, and under Q_SAFE_P_UNSAFE it means UNSAFE. n_{race} is the number of independent races in the cell and n_{dec} the number of decisions recorded in it; the UNSAFE rate is decision-weighted. All twelve cells are complete and no response failed to parse.

Narrative skin	Mapping	Risk	n_{race}	n_{dec}	UNSAFE actions	UNSAFE rate
abstract_game	P_SAFE_Q_UNSAFE	0.1	10	186	186	100.0%
abstract_game	P_SAFE_Q_UNSAFE	0.6	10	186	156	83.9%
abstract_game	P_SAFE_Q_UNSAFE	0.9	10	186	111	59.7%
abstract_game	Q_SAFE_P_UNSAFE	0.1	10	186	186	100.0%
abstract_game	Q_SAFE_P_UNSAFE	0.6	10	186	186	100.0%
abstract_game	Q_SAFE_P_UNSAFE	0.9	10	186	166	89.2%
technology_race	P_SAFE_Q_UNSAFE	0.1	10	186	186	100.0%
technology_race	P_SAFE_Q_UNSAFE	0.6	10	186	144	77.4%
technology_race	P_SAFE_Q_UNSAFE	0.9	10	186	81	43.5%
technology_race	Q_SAFE_P_UNSAFE	0.1	10	186	186	100.0%
technology_race	Q_SAFE_P_UNSAFE	0.6	10	186	185	99.5%
technology_race	Q_SAFE_P_UNSAFE	0.9	10	186	83	44.6%

inverse is not an unresponsive route, since all nine routes move the right way with risk and Claude Opus 5 most steeply of all; eight of them merely also land inside the model’s own range, which is what lets the inversion answer at all. And every curve behind it is the small-mutation limit, which cannot represent the finite mutation rate of the parameter points it is drawn at.

S4 Presentation robustness

S4.1 Crossed representation-robustness design

An earlier local pilot varied the story the game is told through and the symbols used for the two actions, but varied them together with repetition parity, so the two could not be separated, and it failed its screen. It is therefore excluded, and a fully crossed rerun was required before any claim about representation. That rerun has now completed on the two routes selected for this study, and this subsection reports it.

The rerun crosses two *narrative skins*, that is two stories that leave the rules untouched, against two *symbol mappings*, that is which of the two opaque response codes P and Q stands for SAFE and which for UNSAFE, inside repetition blocks, at each of the mechanism’s three maximum-private-risk levels. That is twelve cells per route. It completed on both selected routes under one frozen protocol, the context-and-mapping protocol, version 3, each with ten repetitions, 120 races, 2,232 recorded decisions and zero parse failures. The two routes are Gemini 3 Flash and Claude Sonnet 5, whose full route strings are given in Table 2. The horizon and set-back random-number streams are held common across conditions, so two cells differ in their prompt and not in the environment they met. Table 6 gives all twelve cells for the first route; the second route’s cells are released with the analysis code, and both routes’ paired contrasts are in Table 7.

Pooling cells is not the sharpest way to read this design. The repetition index is a common-random-number block, so the same

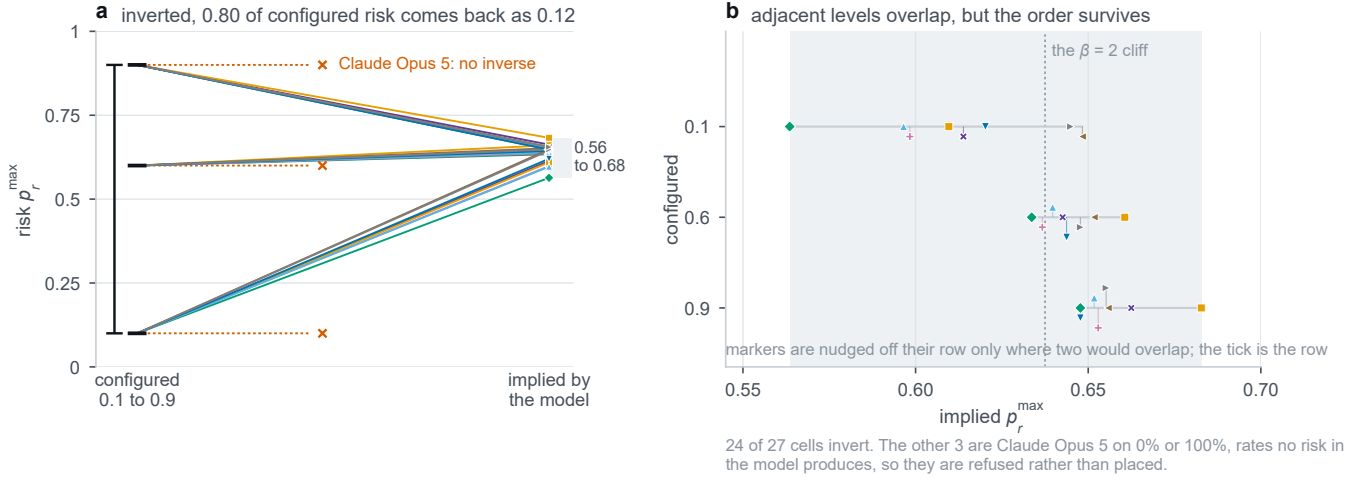


Figure 4: Inverting the reconstructed evolutionary model through the observed UNSAFE rates of the nine audited routes, at the weak-selection point $\beta = 0.01$. (a) The inversion as a slope chart: configured risk on the left at full scale, the risk the model would need in order to emit the route’s observed rate on the right at the same full scale. The fan closes: 0.80 of configured risk comes back as 0.12. Claude Opus 5 has no inverse at any level, because its observed rates of 0% and 100% are rates no risk in the model produces; those three cells are refused rather than placed, and marked as refusals. (b) That band magnified, one row per configured level, with the $\beta = 2$ cliff marked. Adjacent rows overlap, on 0.0147 and 0.0129 of the axis, while the whole configured span of 0.80 collapses to 0.12; what the inversion loses is resolution rather than ranking, since all eight invertible routes keep the configured order and seven of the eight 0.1 cells lie strictly left of every 0.9 cell. Where two markers would fall on each other the second is nudged off its row with a leader back to it; the tick is the row. Neither panel is an estimate of a route’s perceived risk, and a compressed inverse is not an unresponsive route.

index reuses the same sampled horizon across every skin and mapping cell, and a recorded hash of the horizon draws lets that be checked rather than assumed. It holds for all 60 paired blocks on both routes. Differencing inside a repetition therefore removes the horizon draw from the comparison and leaves the presentation as the only thing that varies. Table 7 reports those paired contrasts, with intervals from a percentile bootstrap over repetition blocks.

Two things stand out. The symbol mapping moves play by a similar amount on both routes, about eight to ten percentage points, even though the two routes differ by provider. The narrative skin also moves play on both, but by very different amounts, so skin sensitivity is route-specific in a way that code sensitivity is not. All four intervals exclude zero. Nothing in the mechanism differs across any of these cells: the payoff matrix, the progress increments, the hidden horizon law and the risk formula are identical, and both routes had already passed the quality-control screen. Two routes is enough to show the effect replicates and not enough to characterise endpoints in general.

The crossed table keeps both routes’ cells on one common vertical span. What the table adds here is the question a pooled rate cannot answer: the two factors do not interact on either route, so the narrative skin and the symbol mapping simply add, and the crossing that this rerun was built to test returns an interval spanning zero on both routes.

Both routes’ marginal cells and both paired contrasts are regenerated by the released analysis code, which refuses a route whose race count, ten-per-cell balance or parse-failure count does not check out. It writes both routes’ cell rates with a provenance record

Table 7: Repetition-paired presentation contrasts on the two crossed routes. Each contrast is the mean difference in UNSAFE rate within a repetition block, which holds the sampled horizon fixed; positive means the second level raises UNSAFE play. Intervals are 95% percentile bootstrap intervals over the 60 paired blocks. Horizon pairing was verified for 60 of 60 blocks in every row. Rates pool decisions within a cell, while a contrast is the mean over matched blocks of the within-block difference in race-level rates, so a contrast need not equal the difference of the rates in Table 6.

Route	Contrast	Effect	95% CI
Gemini 3 Flash	code Q means SAFE	+9.5 pp	+5.6 to +13.8
Claude Sonnet 5	code Q means SAFE	+8.3 pp	+5.4 to +12.2
Gemini 3 Flash	abstract contest	+10.8 pp	+6.0 to +15.6
Claude Sonnet 5	abstract contest	+3.4 pp	+0.9 to +7.4

beside them. An earlier attempt on the Claude route stopped at 106 of 120 races on a provider quota refusal. It is retained as a failure record that points to the complete rerun reported here, and none of its cells is reported, because a counterbalanced design read at 106 of 120 races is unbalanced rather than merely smaller.

Two things are visible in Table 6 and neither is a general finding. First, the risk response survives both manipulations: in every one of the four skin-by-mapping combinations the UNSAFE rate is 100% at risk 0.1 and lower at risk 0.9. Second, the mapping is not inert.

Holding the skin fixed, swapping which code means *SAFE* moves the rate at risk 0.9 from 59.7% to 89.2% under the abstract story and from 43.5% to 44.6% under the technology story, and at risk 0.6 from 83.9% to 100% and from 77.4% to 99.5% respectively. A relabelling that a correct model of the game should ignore therefore changes the action sequence.

What follows from this table is bounded by how many routes stand behind it. The design is crossed on both routes selected for the crossed representation study, Gemini 3 Flash and Claude Sonnet 5, whose full route strings are in Table 2, each at the full twelve cells, so the mapping effect is not confined to one provider’s decoder at one revision: it appears on both, at +9.5 and +8.3 points. Two routes are still two routes. They do not separate what belongs to this task from what belongs to commercial endpoints in general, they are not a sample from which a population of models could be described, and the narrative skin already behaves differently on the two, so a third route could as easily break the pattern as extend it. We therefore report the symbol-mapping effect as replicated across the two audited routes, and make no claim about symbol mapping or narrative skin beyond them.

S4.2 Human-benchmark replication check

Before comparing model-generated behaviour with the human benchmark of Fernández Domingos and Han [2] in the main paper’s diversity comparison, we first validated our reconstruction of that benchmark by refitting its published dynamic specification on the de-identified dataset of Fernández Domingos and Han [2]. The reconstructed estimation sample contains 2,888 round-level observations, 172 race-pair clusters, and 338 participants. Table 8 compares the refit coefficients against the values reported in the original study.

The human reference flow is 341 released participants, 340 with a complete first-five-round action history, and 338 with complete values on every covariate in the published dynamic specification. The raw sequence comparison therefore uses 340 participants; the descriptor and elicitation analyses use 341 because they can be computed for one further participant whose complete sequence is unavailable; and the dynamic refit uses 338. No participant is added anywhere, and each figure is a subset of the same released dataset.

Table 8: Replication of Fernández Domingos and Han [2]’s dynamic specification on the de-identified dataset of Fernández Domingos and Han [2]. “Reconstructed” is our refit; “Original” is the coefficient reported in the source study.

Term	Reconstructed	Original
Opponent’s previous action	0.606	0.607
Player’s progress gap	−0.295	−0.296
First-round action	0.217	0.217
Player’s own previous action	−0.195	−0.193

S5 Scripted-opponent results

S5.1 Five audited routes against a fixed and undisclosed rival policy

Every gameplay result reported elsewhere in this paper is self-play: both companies in a race are the same endpoint. That design cannot separate a route responding to its rival from a route running through a phase of its own, because the rival is the route. It also leaves this paper’s two lanes disconnected. The evolutionary lane is built on four reduced strategies, *Always Safe*, *Always Unsafe*, *Conditionally Safe* and *Conditionally Antisocial Safe*, and until this campaign the empirical lane had never played against any of them.

What was collected. Five gate-admitted routes, Gemini 3 Flash, Claude Opus 5, GPT-5.4, GPT-5.5 and Claude Sonnet 5, whose full route identifiers are in Table 2, each played the same game against each of those four strategies at each of the three private-risk levels. That is 60 cells, ten repetitions each, 600 races and 5,580 route decisions, 93 route decisions per cell, with zero parse failures and zero deviations by any scripted rival. The rival is executed by the task file rather than called, so it costs no requests and cannot be wrong about its own strategy for any reason a model could be.

Playing a language model against fixed strategies is established practice rather than something introduced here: Akata et al. [1] run always-cooperate, always-defect and defect-once agents against five models, and Payne and Alloui-Cros [5] run Tit-for-Tat, Suspicious Tit-for-Tat and Grim Trigger, among other strategies, against frontier models under a varying termination probability. Our two conditional rivals are their Tit-for-Tat and Suspicious Tit-for-Tat. What is particular to this campaign is that the rival is never disclosed, that its every move is replayed and checked after the fact, that the route’s seat is counterbalanced, and that the same frozen design is run on more than one endpoint.

Three properties of the design are load-bearing and were asserted offline before any of it was pushed. The prompt is byte-identical to the neutral baseline’s and never says the rival is scripted, so a race here is comparable with a race there; the rival is a strategy, not a disclosure. The route’s seat is counterbalanced, five repetitions in each seat per cell, because this study has measured a seat effect in the neutral baseline, on prompts that differ only in which of two names is whose, and a fixed seat would fold that effect into every number. And a conditional strategy answers the *route*’s previous move rather than its own, which is the property that separates a rival from a mirror: a strategy answering its own history would be a constant, and the whole conditional arm would measure nothing.

How the rival was audited. A scripted rival is code, so it cannot be audited by reading a response. It was audited by replay: the released ingestion code recomputes every rival move from the route’s own recorded moves and refuses the cell on a single deviation. All 60 cells replayed clean. This is the one failure that would have left the data looking perfectly healthy while answering a different question than the one asked.

The collection record. The workload was partitioned into the analysis unit itself, one route by rival by risk cell, and each cell was collected whole on one declared account. The assignment was fixed before the final extension began and is recorded in collection

receipts that are kept outside the paper, because they name the accounts. All 60 cells carry the same scripted-rival protocol and prompt version; the three task hashes are one per risk level and are identical across routes. This keeps the game mechanism fixed while allowing account rotation across cells. The account is a billing boundary, not a model boundary: the analysis never treats account identity as evidence about behaviour, and failed quota or authentication attempts contributed no races to the admitted set. The complete assignment audit is kept with those receipts. The full route-by-risk record is in Tables 9–11, which separate rates, paired contrasts and interactions for readability; the main paper’s scripted-rival figure is the only visualization retained in the evidence package.

Coverage and decoding contracts. The final campaign is complete: all five admitted routes have all 12 route-by-strategy-by-risk cells. Every cell has ten races, 93 route decisions, zero parse failures, zero rival replay deviations and five races in each route seat. The routes are not homogeneous in decoding contract. Gemini 3 Flash, GPT-5.4 and Claude Sonnet 5 use the legacy 256-token, no-reasoning contract; GPT-5.5 uses a 1024-token, low-reasoning contract; Claude Opus 5 uses the latter contract in eleven cells and the legacy contract in one. Coverage therefore supports the five-route descriptive comparison, but not a claim that all endpoints were sampled under one identical decoding configuration. The 36 legacy collection receipts also omit a copied source hash field; their run manifests carry the correct risk-specific hashes, so this is a provenance metadata limitation rather than a gameplay failure.

One-cell contract sensitivity. The one Claude Opus 5 cell collected under the legacy decoding contract is *Always Safe* at risk 0.6. A predeclared harmonisation attempt under the active contract returned a quota 403 before any race or model output was produced, so it is retained as an infrastructure failure and the existing cell is not replaced. Excluding it as a read-only scope check removes one *Always Unsafe* – *Always Safe* contrast and one four-strategy ordering cell. The remaining evidence still has positive rival contrasts in 14/14 cells and the weak ordering in 13/14 cells (strict ordering in 11/14). This is a scope sensitivity, not a harmonised replication; full-grid values remain reported under their recorded contracts.

Finding one: the shape is largely shared, with one boundary reversal. Every route generally plays *UNSAFE* more often as the scripted rival becomes more *UNSAFE*, and the *Always-Unsafe* minus *Always-Safe* contrast is positive in all fifteen route-by-risk cells. The smallest interval lower bound is +16.5 points on Claude Opus 5 at risk 0.1. The weak ordering $\text{Always Safe} \leq \text{Conditionally Safe} \leq \text{Conditionally Antisocial Safe} \leq \text{Always Unsafe}$ holds in fourteen of fifteen cells and is strict in twelve. The one reversal is GPT-5.5 at risk 0.1, where *Conditionally Antisocial Safe* reaches 100.0% and *Always Unsafe* reaches 97.8%. Both values are close to the ceiling, so the reversal is a saturated boundary observation rather than evidence against the broader direction. Table 10 reports every paired contrast used in this summary.

The response to stated risk survives once the rival is held fixed. Against *Always Safe*, the rate never rises with risk on any of the five routes. The paired risk contrast ranges from +10.4 to +69.9 points, with Claude Opus 5 providing the largest change and the other four routes showing more graded changes. This contrast is

not interchangeable with the rival contrast: both are paired within repetitions, but they change different parts of the interaction.

Finding two: risk changes the rival effect in endpoint-specific directions. Table 11 tests the interaction directly as the high-risk minus low-risk change in the paired *Always-Unsafe* minus *Always-Safe* contrast. The interaction is positive on Claude Opus 5 (+29.2 points [20.9, 36.4]) and GPT-5.5 (+36.4 [29.0, 44.0]), negative on Claude Sonnet 5 (−18.9 [−25.1, −11.8]), and unresolved on Gemini 3 Flash and GPT-5.4 because both intervals contain zero. The data therefore reject a single claim that risk either amplifies or separates the rival effect across endpoints. These are within-endpoint interactions; cross-endpoint effect sizes remain descriptive because decoding contracts differ.

We name the contrast for what it compares. Calling the safe-rival arm exploitation would assert that the route takes the opportunity a non-punishing rival offers, and the measured rates say the opposite: on every route that arm carries the lowest rates anywhere in the campaign, below what the same route plays against itself. What the routes do is match the rival, keeping most of their restraint against a rival that keeps its own and abandoning much of it against one that never had any. A strategy name is not a summary of what the rival played either. *Conditionally Safe* opens *SAFE* and then mirrors; its own *Unsafe* rate varies with the route because it follows the route’s previous move. Across the five routes it played *UNSAFE* on approximately 31% to 73% of its own moves, so the strategy label should not be read as a claim about its realised average action.

The rival’s opening move. *Conditionally Safe* and *Conditionally Antisocial Safe* are the same strategy from round two onward, both repeating the route’s previous move, so their difference isolates the rival’s first move and nothing else. The full opening-move contrast is retained in the derived route tables. It is not used as the headline result because its interpretation is conditional on the route’s own first move, which varies substantially across the five routes.

The route’s own opening move is not a constant across the campaign: Gemini 3 Flash opened *UNSAFE* in all 120 races, Claude Opus 5 in 41, GPT-5.4 in 116, GPT-5.5 in 115, and Claude Sonnet 5 in 23. The opening-move contrast is about the rival’s first move, which the task file fixes, and not about the route’s.

What it means for the rest of this paper. A self-play rate is not a measurement of how a route treats risk. Self-play exceeds the fixed-safe rate in thirteen of fifteen cells. The two exceptions are Claude Opus 5 at the floor. At risk 0.6 its self-play rate is lower (0.0% against 1.1%), and at risk 0.9 the two designs are tied at 0.0%. The risk-0.1 Gemini self-play distance is a lower bound, because nine of its ten mirror-match races have every decision *UNSAFE* and that arm has no room above it. A self-play rate is an interaction outcome of one policy meeting a copy of itself, and every self-play number in this paper should be read that way.

That self-play can mask behaviour is not a new observation. Lekeas and Stamatopoulos [4] report phenomena that are, in their words, invisible in self-play, including that which agent moves first decides which equilibrium a pair reaches. The contribution here is narrower and specific to this setting: self-play can alter or mask the magnitude and shape of the *risk response*, the quantity this game

Table 9: Rates in the full five-route scripted-rival campaign. Each AS/CS/CAS/AU entry is one route-by-strategy-by-risk cell with ten races and 93 route decisions. Both seats are used in every cell, with zero parse failures and zero rival replay deviations. Rates are percentages of the route’s own decisions; self-play is the corresponding neutral-baseline rate.

Endpoint	$p_r^{\max}$	AS	CS	CAS	AU	Self-play
Gemini 3 Flash	0.1	24.7	76.3	100.0	100.0	98.9
	0.6	17.2	57.0	74.2	89.2	73.1
	0.9	14.0	57.0	67.7	88.2	60.2
Claude Opus 5	0.1	63.4	100.0	100.0	100.0	100.0
	0.6	1.1	2.2	53.8	89.2	0.0
	0.9	0.0	3.2	35.5	62.4	0.0
GPT-5.4	0.1	36.6	72.0	86.0	89.2	87.1
	0.6	32.3	55.9	69.9	83.9	65.1
	0.9	25.8	50.5	62.4	80.6	46.8
GPT-5.5	0.1	72.0	98.9	100.0	97.8	93.5
	0.6	23.7	82.8	87.1	91.4	70.4
	0.9	21.5	66.7	75.3	87.1	55.4
Claude Sonnet 5	0.1	22.6	63.4	89.2	90.3	89.2
	0.6	10.8	43.0	62.4	77.4	48.4
	0.9	10.8	44.1	47.3	57.0	32.3

Table 10: Paired contrasts underlying the scripted-rival result. Rival stance is Always Unsafe minus Always Safe at one risk level. Risk at Safe is risk 0.1 minus risk 0.9 with the rival held at Always Safe. Each entry is a percentage-point mean with a 95% percentile interval over ten matched repetition blocks. Contrasts are within endpoints; cross-endpoint magnitudes are descriptive because decoding contracts differ.

Endpoint	Rival stance at stated risk			Risk at Safe 0.1 – 0.9
	0.1	0.6	0.9	
Gemini 3 Flash	+73.5 [67.8, 78.7]	+67.4 [57.1, 76.6]	+71.7 [65.6, 77.5]	+10.4 [5.4, 15.7]
Claude Opus 5	+30.1 [16.5, 42.8]	+87.0 [84.3, 89.6]	+59.3 [51.2, 66.5]	+69.9 [57.1, 83.5]
GPT-5.4	+47.5 [37.0, 57.2]	+48.0 [40.0, 55.0]	+49.8 [35.8, 60.5]	+12.3 [6.2, 18.6]
GPT-5.5	+25.3 [20.3, 30.6]	+66.1 [62.5, 69.9]	+61.7 [53.5, 69.1]	+48.3 [43.1, 54.3]
Claude Sonnet 5	+63.8 [54.8, 72.1]	+65.7 [62.2, 68.9]	+45.0 [42.0, 48.2]	+12.7 [8.5, 16.6]

Table 11: Paired rival-by-risk interaction, $[(AU-AS)_{0.9} - (AU-AS)_{0.1}]$, in percentage points. Positive values mean that the rival contrast grows from low to high risk.

Endpoint	Mean [95% CI]
Gemini 3 Flash	−1.8 [−7.9, 3.7]
Claude Opus 5	+29.2 [20.9, 36.4]
GPT-5.4	+2.3 [−4.6, 9.4]
GPT-5.5	+36.4 [29.0, 44.0]
Claude Sonnet 5	−18.9 [−25.1, −11.8]

exists to elicit. Every route in the scripted-rival campaign changes behaviour against at least one fixed rival.

What it does not show. Five routes, one game, one prompt version, ten races per cell. Five commercial endpoints are not a sample of a population of models, and four strategies are four points in a space of policies rather than a sample of rivals, so a scripted rival establishes what a route does against *these* rivals. The within-block policy contrast is identified because rival-policy assignment is fixed and exogenous. Conditional rivals can still respond mechanically to the route’s previous action, as specified by their fixed policies. This is not causal about anything outside this game. Nothing here licenses a claim about strategy, intent, or reasoning: these are conditional frequencies. It is regenerated by the released analysis code,

which takes one endpoint at a time, writes one derived table per endpoint rather than pooling them, and refuses a cell that did not complete, that carries a parse failure, that is the wrong protocol, that has the wrong number of races, whose seat counterbalance is not five and five, or whose rival deviated from the strategy its directory claims.

S6 Human reference and trajectory-diversity robustness

S6.1 The pilot roster behind the projections

The diversity comparison in the main paper reads all nine audited routes on the confirmatory baseline. The population classifier and two-dimensional embedding in this appendix instead use an earlier seven-checkpoint export of the same two-player game. That roster is used only for those exploratory trajectory analyses; it does not redefine the nine-route human-diversity roster in the main paper. It pools 420 trajectories, 60 per checkpoint, with 340 complete human trajectories from the de-identified dataset of Fernández Domingos and Han [2], one incomplete human record excluded, $N = 760$. Six of the seven checkpoints carry an endpoint-admission verdict from §S1.1; GPT-5 nano is no longer offered on the audited account and so cannot be gated at all, and is retained as description rather than as evidence. Each player is represented by 15 raw features: own

action, opponent action, and own-minus-opponent progress gap entering each of rounds 1 to 5, which are the recorded states and actions themselves rather than derived quantities.

No single aggregate rate characterises these checkpoints. Mean UNSAFE play over rounds 1 to 5 is lowest for GPT-5 nano (17%), then Claude Sonnet 5 (22%), Claude Opus 5 (33%), GPT-5.4 nano (54%) and humans (56%), with the Gemini checkpoints running from 73% (Gemini 3 Flash) to 83% (Gemini 3.1 Flash Lite). Closeness to the human mean is, on its own, uninformative about anything beyond that one number, which is why the main paper measures diversity on the action sequences instead.

The two-dimensional embedding of the same space is §S6.4, and neither the main paper nor this appendix rests a conclusion on it: recolouring its identical coordinates by outcome rather than by population separates it almost entirely by UNSAFE rate.

The heterogeneity those pictures suggest is also testable directly, and the test belongs here because it covers five checkpoints rather than the nine the main paper reads. Fitting three nested logistic models to the pooled neutral-baseline pilot, risk only, checkpoint identity only, and checkpoint identity by risk, shows that letting the risk-response profile vary by checkpoint improves fit substantially beyond checkpoint-specific intercepts alone, $\chi^2(10) = 354.7$, $p < 10^{-3}$, with several checkpoint-by-risk cells approaching complete separation. A convergence warning in the interaction model means the displayed likelihood-ratio statistic is approximate, so only the conservative threshold is quoted. The fit covers the five-checkpoint pilot baseline only, and the two Claude checkpoints are not in it. Neither caveat changes what it supports: no single risk-response curve represents all five, so the differences between checkpoints are differences in *shape* and not only in level.

S6.2 Population identity in first-five-round trajectories

Table 12 is a supplementary diagnostic: it asks how well a simple classifier can name the population that produced a trajectory, using the same 15 raw features as the trajectory embedding of §S6.4. We fit a shallow decision tree with class-balanced loss under stratified five-fold grouped cross-validation. The two seats of an LLM race remain in the same fold, and human trajectories are grouped by source dyad. Since the pooled classes are unequal, balanced accuracy and macro-F1 are the primary measures. A 200-repetition permutation null shuffles one population label per race or dyad within equal-size strata, preserves the observed class counts and grouped folds, and records one mean across the five folds per permutation.

Both primary scores exceed chance (1/8) and the grouped permutation null (both $p = 0.005$). Raw accuracy is also above its grouped null ($p = 0.005$), but we report it descriptively because the human class is larger. The result supports observable differences in the first-five-round trajectories, while the overlap and the shallow classifier do not support a claim that a latent strategy or model identity is cleanly recoverable. The machine-readable result and its source hashes are released with the analysis code.

S6.3 Predictive structure of UNSAFE choices

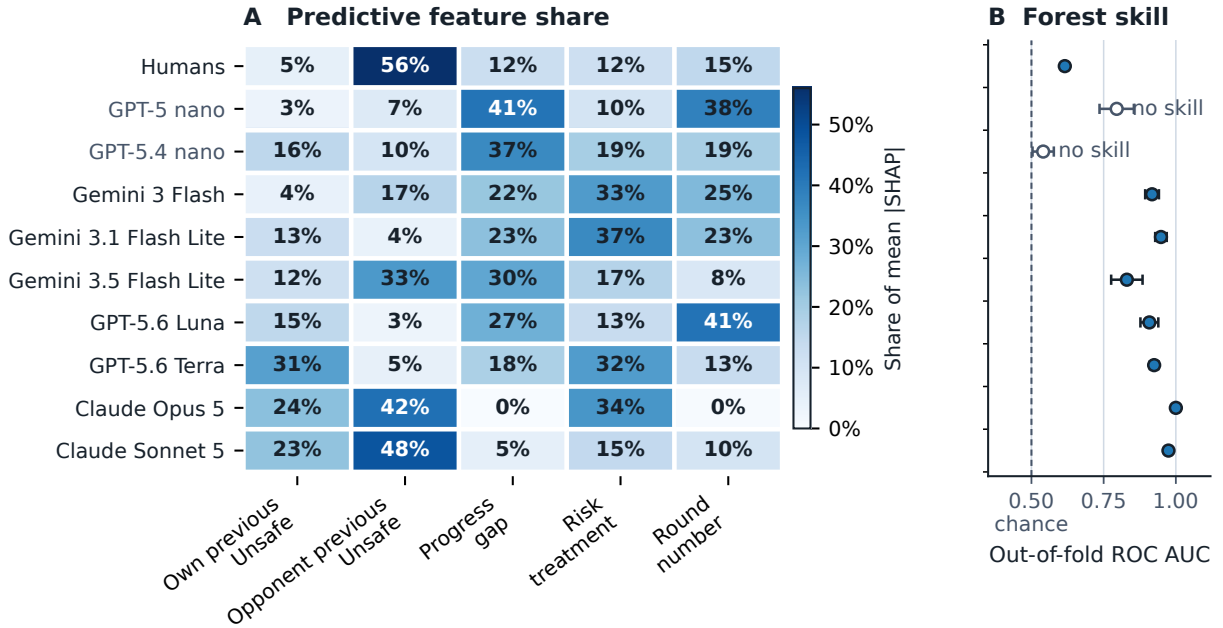
The preceding section shows that populations produce different action patterns. The predictive analysis uses a separate exploratory

Table 12: Dyad/race-grouped population-identity diagnostic on 760 complete first-five-round trajectories. The human population has 340 trajectories and each of the seven listed checkpoints has 60; the classes are therefore imbalanced, so balanced accuracy and macro-F1 are primary. The null interval is the 2.5–97.5 percentile range over grouped label permutations.

Metric	Observed	Permutation null mean	Null 95% interval
Raw accuracy	0.282 ± 0.073	0.088	0.066–0.120
Balanced accuracy	0.384 ± 0.057	0.127	0.092–0.168
Macro-F1	0.237 ± 0.051	0.064	0.040–0.092

roster: nine checkpoint-specific neutral-lane datasets, with 30 or 60 races per checkpoint, plus the human decisions. Table 13 lists that roster; it is not the seven-checkpoint trajectory sample above. We ask which recorded parts of the game best predict their next UNSAFE choice. This is a study of association, not cause or internal reasoning. For each population, we fit a random forest, a model that combines many decision trees. Table 13 reports the result and Figure 5 shows the same shares as a heatmap. Predictive scores use five-fold grouped cross-validation, with all decisions from one source dyad or race kept in one fold. The forest predicts decisions after round one from five values known before the choice: the player’s previous action, the opponent’s previous action, the progress gap, the assigned maximum private risk, and the round number. Tree-SHAP then measures how much each value contributes to the fitted model’s predictions. A larger SHAP share means that the predictor relied more on that feature. It does not mean that the feature caused the decision.

For human participants, the opponent’s previous action accounts for 56% of total SHAP magnitude, far more than any other feature, while the player’s own previous action contributes only 5%: a reciprocity-dominated profile that independently recovers the pattern found in the human logistic analysis. One caveat has to travel with that reference profile everywhere it is used below, because it is the yardstick the checkpoints are held against and it is the least well identified row in the table. The human forest barely discriminates: ROC AUC 0.62, balanced accuracy 0.57, and a test accuracy of 0.608 against a majority-class baseline of 0.588, so it is about 2.0 points of accuracy above predicting the more common action and nothing else. Adding the demographic covariates does not help, AUC 0.61. The model-side forests are far better fitted, up to AUC 1.00 for Claude Opus 5 and 0.97 for Claude Sonnet 5, so every comparison in this subsection is between shares extracted from very differently identified models. That is not a reason to drop the analysis, and it is a reason not to read a matched ordering as a matched policy. GPT-5-nano is close to the opposite profile: relative race position (progress gap) accounts for 41% of its predictive importance and opponent history for only 7%, and its apparently strong out-of-fold AUC of 0.80 is misleading on its own, since balanced accuracy is exactly 0.50: the checkpoint plays so close to the SAFE floor that the classifier effectively fails to recover the rare UNSAFE class. Gemini-3-Flash and Gemini-3.1-Flash-Lite are primarily risk-treatment-driven (33% and 37% of SHAP magnitude), consistent with their strong, monotone descriptive risk-response



Associational feature use by separately fitted forests; it is not a causal attribution. Panel A normalises every row to 100% whether or not the forest predicts anything, so panel B carries the skill each row rests on: whiskers are the cross-validated standard deviation, and an open marker flags a forest whose balanced accuracy is at chance and whose accuracy does not beat the majority class.

Figure 5: Population-specific predictive structure for round- $t \geq 2$ UNSAFE choices. (A) Each row reports the share of mean absolute SHAP value assigned to the same five pre-decision features by a random forest fitted separately to that population. SHAP magnitudes describe how the fitted classifier uses observed features; they are neither causal effects nor evidence of the agents’ internal reasoning. (B) The out-of-fold discrimination each row rests on, because panel A normalises every row to 100% whether or not its forest predicts anything; whiskers are the cross-validated standard deviation and an open marker flags a forest at chance. Read the two panels together, because they are not on the same footing. The human forest discriminates weakly, ROC AUC 0.62 and balanced accuracy 0.57, about 2.0 points of accuracy above guessing the majority class, while the two Claude forests reach 1.00 and 0.97. A share from a weakly fitted forest says mostly where that forest found the little signal there was, so it is a much looser proxy for the player’s policy than a share from a well fitted one.

Table 13: Population-specific random-forest prediction of UNSAFE play. Feature columns report shares of total mean absolute SHAP magnitude (rows sum to 100%, up to rounding); AUC and balanced accuracy are mean out-of-fold values from the population-specific race/dyad-grouped five-fold split.

Population	Own previous	Opponent previous	Progress gap	Risk	Round	OOF AUC	Balanced accuracy
Human	5%	56%	12%	12%	15%	0.62	0.57
GPT-5-nano	3%	7%	41%	10%	38%	0.80	0.50
GPT-5.4-nano	16%	10%	37%	19%	19%	0.54	0.52
Gemini-3-Flash	4%	17%	22%	33%	25%	0.92	0.77
Gemini-3.1-Flash-Lite	13%	4%	23%	37%	23%	0.95	0.75
Gemini-3.5-Flash-Lite	12%	33%	30%	17%	8%	0.83	0.71
GPT-5.6-Luna	15%	3%	27%	13%	41%	0.91	0.78
GPT-5.6-Terra	31%	5%	18%	32%	13%	0.92	0.85
Claude Opus 5	24%	42% [†]	0%	34%	0%	1.00	1.00
Claude Sonnet 5	23%	48%	5%	15%	10%	0.97	0.93

[†] Collinear with the risk treatment, not a reciprocity estimate; see the discussion below.

curves. Claude Sonnet 5 is the checkpoint closest to the human profile, and it is the exception to any claim that no model shares it: the opponent’s previous action is its largest predictor too, at 48% against the human 56%. Be exact about how far that goes. Two

other checkpoints also rank that feature first, Claude Opus 5 at 42%, which the dagger above marks as collinear with the risk treatment rather than a reciprocity estimate, and Gemini-3.5-Flash-Lite at 33% with its progress gap close behind at 30%. And beyond the

leading feature no checkpoint reproduces the human rank order: below the opponent term the human profile runs round number, progress gap, risk, own previous, while Claude Sonnet 5 runs own previous, risk, round number, progress gap. The agreement is in which feature leads, not in the ordering as a whole. GPT-5.4-nano has no comparably dominant predictor: importance is spread across progress gap, risk, round, and its own previous action, with opponent history at 10% and the lowest above-chance predictive fit in the table (AUC 0.54, balanced accuracy 0.52), so its behaviour is comparatively difficult to reconstruct from these five mechanical state variables rather than clearly governed by any one signal.

Claude Opus 5 shows why these shares must be read against the design that produced them. Its 42% opponent share reads as strong reciprocity and is not: on the neutral lane it plays UNSAFE on essentially every low-risk decision and essentially none at high risk, so the risk treatment alone separates its choices, and since both seats behave that way the opponent’s previous action is a proxy for the risk level rather than an independent signal. The forest then splits importance between two variables encoding the same event, and the perfect AUC describes a step function, not a better fit. A feature-importance share is interpretable only where the feature keeps variation independent of the others, which is what a near-deterministic policy removes.

One marginal association does agree across populations: human winners play UNSAFE 16.0 percentage points more often than human losers, against an approximately 20-point pooled difference among the checkpoints. The dynamic structure behind that agreement does not agree. GPT-5.4-nano in particular has no single strong predictor, and several checkpoint estimates are weak, reversed in sign, or not reliably estimable because their action distributions sit at the floor or the ceiling.

Predictive performance varies substantially across populations (AUC 0.54 to 1.00), so SHAP shares should not be read as directly comparable measures of decision quality; floor and ceiling behaviour mechanically limits minority-class prediction for several models. The defensible conclusion is narrower than “LLMs use different features”: the same observable state variables organise predictive information differently across populations, and matching a human aggregate action rate does not imply matching the human reciprocity-dominated association profile.

S6.4 Trajectory embedding recoloured by outcome

The embedding of raw five-round trajectories separates almost entirely by how much UNSAFE play a trajectory carries. Both colourings of it live here. Figure 6 colours it by population, and Figure 7 is the check: the same coordinates, with only the colouring changed, so any structure visible in the second is structure the embedding already had. Reading them in that order is the point. The population panels look like eight different regions; recolouring the identical points by each player’s own UNSAFE rate splits the embedding into a SAFE region, an UNSAFE region and a thin mixed band, with very few points on the wrong side. The apparent separation of populations is therefore largely a separation of action rates, which is why the main paper rests no conclusion on either picture and uses the grouped classifier of Table 12 instead.

S6.5 Player-level UNSAFE rate by population

Figure 8 is the aggregate level of the same comparison: one number per population, the mean share of a player’s first five decisions that were UNSAFE. It is the plainest form of the point that a matching aggregate rate does not imply matching behaviour, because a population can reach a middling rate either by mixing or by every player behaving the same way.

Risk-stratified distributions. The pooled mean above is only a location summary. Figure 9 shows the underlying player-level spread separately at each assigned risk level. It uses every retained historical neutral or no-persona lane with player records, so its checkpoint roster is broader than the seven-checkpoint embedding roster and is separate from the nine-route confirmatory roster. The comparison is descriptive: it asks whether a population reaches its mean by placing players across the range or by concentrating them in a narrow band.

S6.6 Trajectory-diversity statistics

The main paper reports that human first-five-round trajectories are more varied than most admitted route-risk cells under a cluster-aware null. The primary dyad-clustered summary is in Table 14. Table 16 retains the full nine-route table, with participant-level human resampling and the companion Hill and Hamming statistics. Together they make each value checkable against its population, its risk condition, and the independent unit it rests on. Table 15 is the earlier seven-checkpoint pilot export, kept because it carries GPT-5 nano, an endpoint no longer offered on the audited identity and therefore holding no admission verdict. That export has no row for GPT-5.4 or GPT-5.5, which are two of the five admitted routes, so it cannot settle a claim about the admitted set and is not used for one.

A trajectory here is the pair of actions taken by the two players over the first five rounds, that is ten SAFE or UNSAFE bits. Three statistics describe how varied a set of such trajectories is. Hill $q = 0$ counts how many *distinct* trajectories appear, so it treats a trajectory seen once and a trajectory seen forty times alike. Hill $q = 1$ is the effective number of trajectories, the exponential of Shannon entropy, which discounts trajectories that are rare: it equals $q = 0$ when every trajectory is equally common and falls below it when a few dominate. Mean pairwise Hamming distance is the average number of the ten bits on which two trajectories drawn from the set differ, divided by ten, so 0 means every trajectory in the set is identical and 0.5 means two trajectories differ on half their bits on average.

Sample sizes differ across populations, and both counting statistics rise with sample size, so every cell is compared at the same size. The point estimate is the mean over draws of 20 trajectories taken without replacement from the cell, or the exact cell value where the cell already holds exactly 20. For the participant-level sensitivity in Table 15, the interval resamples human participants and checkpoint races with replacement, then rarefies to 20 trajectories inside each of 2,000 resamples. A checkpoint race remains the cluster because its two seats share the same sampled horizon and setback draws. The n_{cluster} column reports participants for the human rows and races for the checkpoint rows. The human-side $q=0$ comparison used for the main figure is instead cluster-aware at the dyad level: it samples ten complete human dyads, retaining twenty trajectories

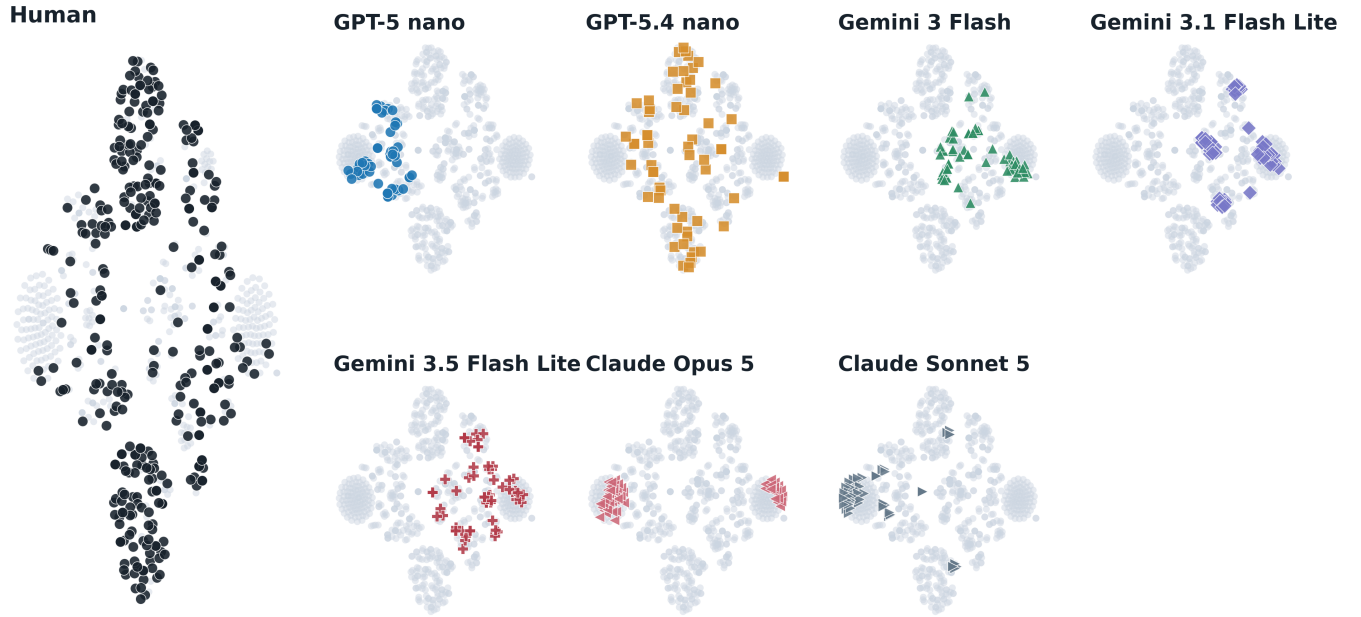


Figure 6: The population-coloured embedding, shown here so that Figure 7 can be compared against it directly. A t-SNE embedding places each of the 760 complete first-five-round trajectories at a point in two dimensions, arranged so that trajectories with similar action-and-position histories fall near each other; the input is the pooled 15-dimensional first-five-round action-and-position space, the same input as the population-identity classifier below. One panel is drawn per population, human first. The axes have no units and no direction: only relative closeness carries meaning, and distances between well-separated clusters are not interpretable. The main paper rests no conclusion on this figure, and the reason is Figure 7: recolouring these identical coordinates by each player’s own UNSAFE rate separates the embedding almost entirely, so what looks like a separation of populations here is largely a separation of action rates. The quantitative statement about population identity is the grouped classifier of Table 12, not this picture.

Table 14: Primary dyad-clustered human null for distinct paired first-five-round trajectories. Each draw samples ten complete dyads, contributing twenty trajectories, without replacement from the risk pool. The mean and 95% RI are over 20,000 draws; the minimum is the finite-draw floor used for the strict below-null comparison. “Admitted below” counts the five screened routes whose confirmatory $q=0$ value is below every draw; “all routes below” uses all nine audited routes.

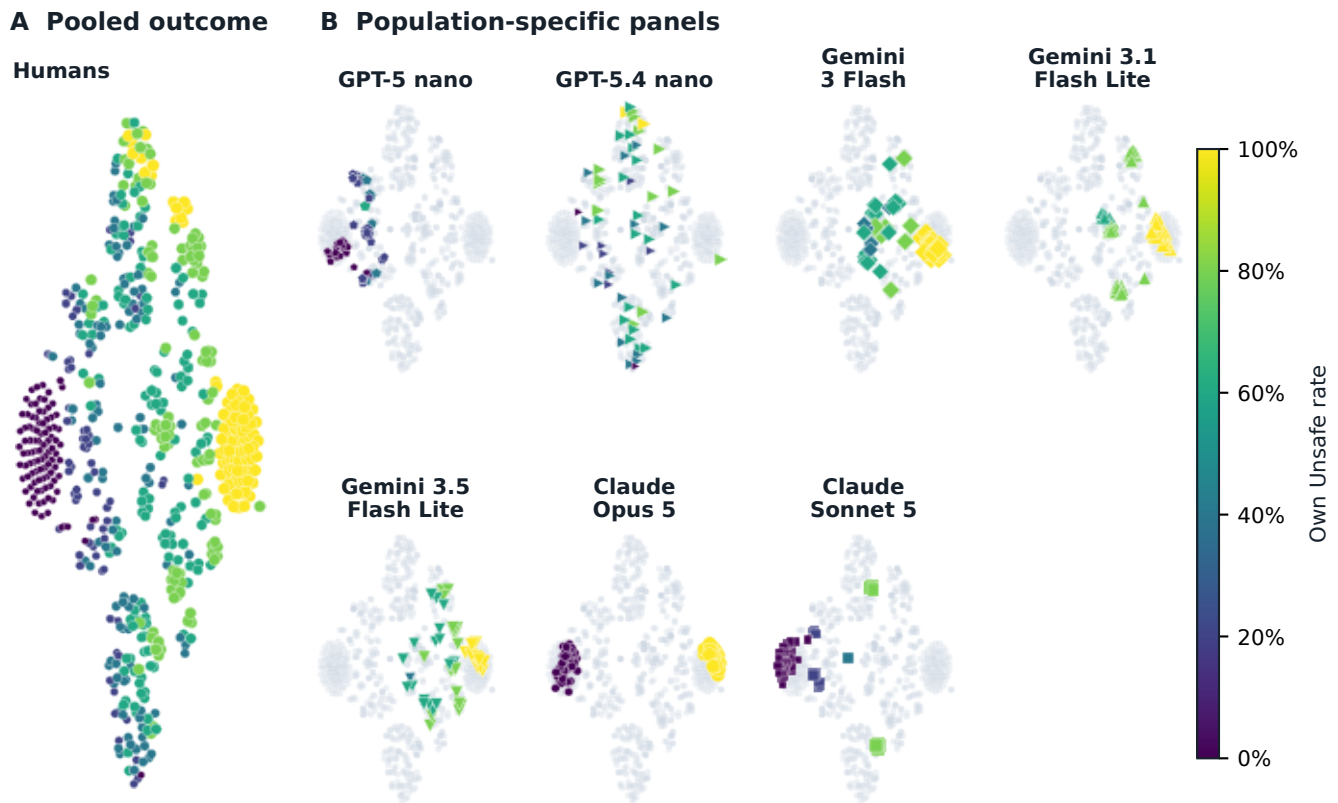
Risk	Dyads	Mean	95% RI	Min	Admitted	All routes
0.1	48	18.6	16–20	12	5/5	7/9
0.6	52	19.6	18–20	15	5/5	7/9
0.9	68	19.2	17–20	13	3/5	4/9

but ten interaction units. Four complete human trajectories have no complete partner in the de-identified file and are omitted from that primary dyad null.

The ordering in Table 15 is consistent across all three risk levels and all three statistics. The human rows sit near the ceiling of the comparison: 19.2, 19.7, and 19.5 distinct trajectories out of a possible 20, with mean pairwise Hamming distance between 0.467 and 0.497, close to the 0.5 that two unrelated trajectories would give.

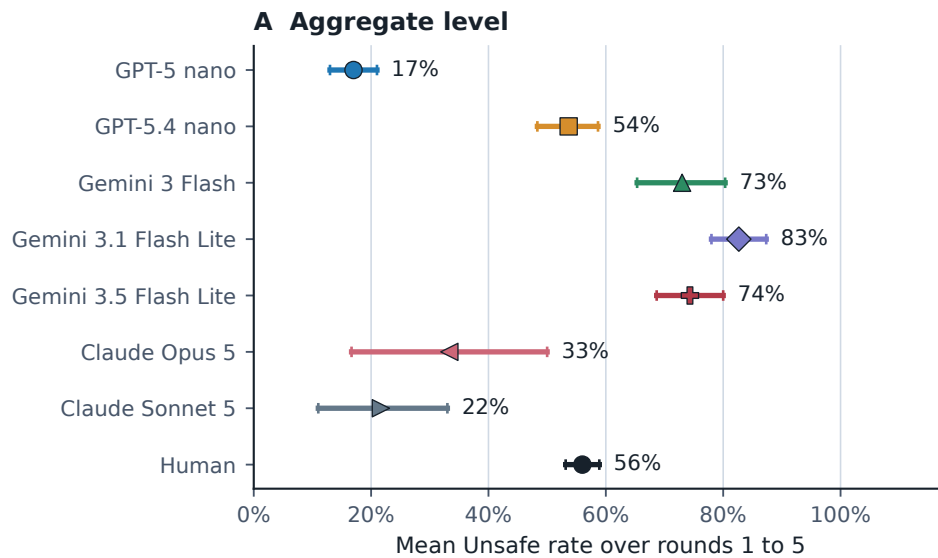
Only one checkpoint in this pilot export reaches that: GPT-5.4 nano returns 19.0 distinct trajectories at every risk level, with Hamming between 0.478 and 0.506. The rest are far below. Claude Opus 5 returns exactly one distinct trajectory at all three risk levels, with Hamming 0, meaning every one of its player-races produced the identical ten-action sequence. Gemini 3 Flash and Gemini 3.1 Flash Lite collapse the same way at risk 0.1. Claude Sonnet 5 falls from 5.0 distinct trajectories at risk 0.1 to 3.0 at risk 0.9, with Hamming 0.020 at the latter. This is the shape of the finding: a checkpoint can match a human aggregate action rate while producing a small handful of distinct action sequences, or only one.

Table 16 is the version of that sensitivity that can be read against the admission gate, because it has a row for every route the gate ruled on. The primary $q=0$ model entries in the table, read against the dyad floors in Table 14, put 13 of 15 admitted cells below every dyad draw. The two admitted boundary cells are GPT-5.4 and GPT-5.5 at risk 0.9; both have 13 or 14 distinct trajectories against a dyad floor of 13. The participant-level Hill $q=1$ sensitivity places every admitted route below the human interval at all three risk levels. GPT-5.4 mini and GPT-5.4 nano remain the two refused routes that reach the participant-level human range at every risk. Claude Opus 5 again returns exactly one distinct trajectory at all three risk levels, with Hamming 0. The dyad correction changes the scope of the $q=0$ claim, not the concentration result itself.



Colour and marker area both rise with the player's own Unsafe rate; marker shape is the route, carried over from the main paper. Panel A pools all 760 trajectories.

Figure 7: The same t-SNE coordinates as Figure 6, recoloured by each player's own mean UNSAFE rate over rounds 1–5 on a monotonic viridis low-to-high rate scale instead of population identity. Left: all 760 trajectories pooled. Right: the same colouring broken out per population.



Points are mean Unsafe rates over player-race trajectories;
 whiskers are 95% race-clustered bootstrap intervals
 (30 races per model, 172 human dyad groups; 4000 resamples).

Figure 8: Player-level mean UNSAFE rate over rounds 1 to 5, one row per population. The point is the unweighted mean over player-race trajectories. The whisker is a 95% percentile bootstrap interval that resamples *clusters*, not single trajectories, because the two seats of a race are not independent of each other: they share the sampled stopping horizon and the setback draws. The cluster is therefore the race for every checkpoint row and the source dyad for the human row, with singleton complete trajectories retained as singleton clusters; the interval uses 4,000 resamples. The point estimates are unchanged from a trajectory-level calculation; only the interval changed, and it supersedes an earlier interval that resampled player-race trajectories and so treated the two seats of a race as independent.

Distributional comparison

Human trajectories are broad; model checkpoints occupy narrower, model-specific bands.

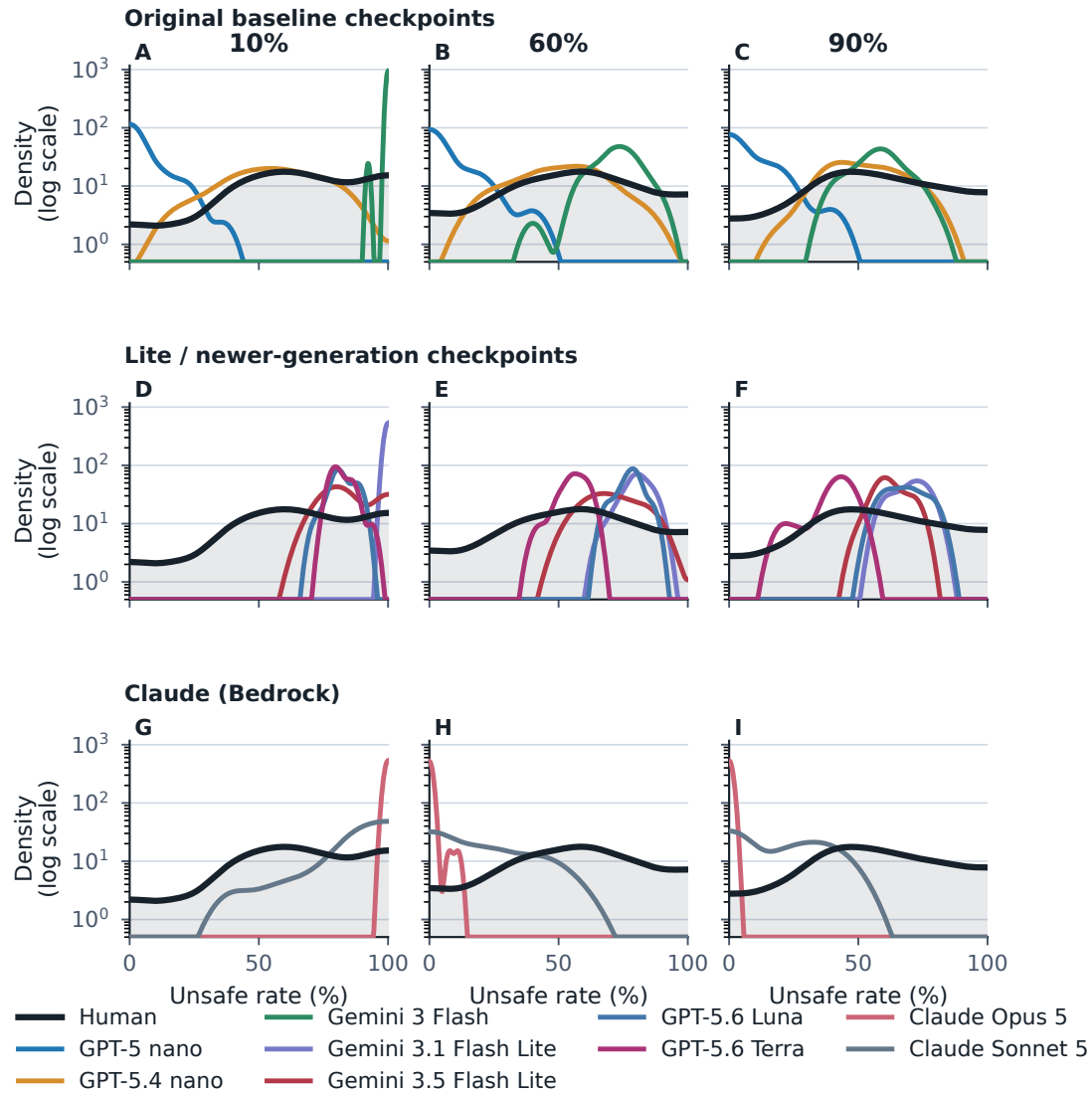


Figure 9: Player-level UNSAFE-rate distributions by assigned risk condition over rounds 1–5. Each curve is a boundary-reflected kernel density on the valid 0–100% range, normalised within population and risk; the vertical axis is logarithmic so near-deterministic checkpoint bands do not flatten the broader human reference. The human curve is repeated in each row, while checkpoint families are split across rows only for legibility. Across all three risk conditions, human players span much of the available range and the plotted checkpoints occupy narrower, checkpoint-specific bands. These historical neutral-lane diagnostics do not pool checkpoints or transfer admission verdicts across rosters.

Table 15: Participant-level sensitivity for sample-size-matched diversity of paired first-five-round trajectories, by maximum-private-risk level and population. A trajectory is the two players’ ten actions over rounds 1 to 5. Hill $q=0$ is the number of distinct trajectories, Hill $q=1$ the effective number after discounting rare ones, and Hamming the mean fraction of the ten bits on which two trajectories differ. n_{src} is the trajectories available in the cell and n_{cluster} the independent units it rests on: participants for the human rows, races for the checkpoint rows. Every cell is compared at 20 trajectories. The parenthesised ranges are 95% *resampling intervals* (RI) and not confidence intervals: they are two-stage cluster bootstrap ranges, with clusters resampled with replacement and then rarefied to 20 within each resample, over 2,000 resamples, and they bound how much the statistic moves under resampling rather than making a confidence statement about the observed count. Because the point estimate is the observed cell value while the interval is computed on resampled clusters, which lose distinct trajectories to duplication, an interval can sit below its point estimate.

Risk	Population	n_{src}	Hill $q=0$ (95% RI)	Hill $q=1$ (95% RI)	Hamming (95% RI)	n_{cluster}
0.1	Human	98	19.2 (15.0–20.0)	18.9 (13.2–20.0)	0.467 (0.407–0.504)	98
	GPT-5 nano	20	13.0 (5.0–12.0)	10.2 (3.4–10.7)	0.275 (0.163–0.339)	10
	GPT-5.4 nano	20	19.0 (9.0–16.0)	18.7 (7.3–15.2)	0.478 (0.405–0.505)	10
	Gemini 3 Flash	20	1.0 (1.0–1.0)	1.0 (1.0–1.0)	0.000 (0.000–0.000)	10
	Gemini 3.1 Flash Lite	20	1.0 (1.0–1.0)	1.0 (1.0–1.0)	0.000 (0.000–0.000)	10
	Gemini 3.5 Flash Lite	20	5.0 (3.0–5.0)	3.6 (1.9–4.8)	0.166 (0.076–0.227)	10
	Claude Opus 5	20	1.0 (1.0–1.0)	1.0 (1.0–1.0)	0.000 (0.000–0.000)	10
	Claude Sonnet 5	20	5.0 (3.0–5.0)	4.5 (2.6–5.0)	0.354 (0.246–0.413)	10
0.6	Human	104	19.7 (15.0–20.0)	19.7 (14.1–20.0)	0.495 (0.455–0.513)	104
	GPT-5 nano	20	8.0 (5.0–8.0)	7.2 (3.4–7.7)	0.244 (0.154–0.301)	10
	GPT-5.4 nano	20	19.0 (9.0–16.0)	18.7 (7.2–15.2)	0.506 (0.402–0.520)	10
	Gemini 3 Flash	20	11.0 (5.0–11.0)	9.3 (4.4–10.2)	0.312 (0.208–0.371)	10
	Gemini 3.1 Flash Lite	20	4.0 (2.0–4.0)	2.8 (2.0–3.7)	0.229 (0.211–0.255)	10
	Gemini 3.5 Flash Lite	20	12.0 (6.0–12.0)	11.5 (5.5–10.9)	0.364 (0.267–0.397)	10
	Claude Opus 5	20	1.0 (1.0–1.0)	1.0 (1.0–1.0)	0.000 (0.000–0.000)	10
	Claude Sonnet 5	20	5.0 (1.0–5.0)	2.7 (1.0–4.0)	0.074 (0.000–0.133)	10
0.9	Human	138	19.5 (16.0–20.0)	19.3 (14.8–20.0)	0.497 (0.457–0.516)	138
	GPT-5 nano	20	12.0 (6.0–12.0)	10.0 (3.9–10.7)	0.232 (0.134–0.304)	10
	GPT-5.4 nano	20	19.0 (9.0–16.0)	18.7 (7.3–15.2)	0.486 (0.394–0.504)	10
	Gemini 3 Flash	20	11.0 (6.0–11.0)	10.7 (5.5–10.2)	0.311 (0.262–0.314)	10
	Gemini 3.1 Flash Lite	20	3.0 (2.0–3.0)	2.9 (2.0–3.0)	0.101 (0.079–0.105)	10
	Gemini 3.5 Flash Lite	20	12.0 (6.0–12.0)	11.5 (5.2–10.9)	0.401 (0.319–0.417)	10
	Claude Opus 5	20	1.0 (1.0–1.0)	1.0 (1.0–1.0)	0.000 (0.000–0.000)	10
	Claude Sonnet 5	20	3.0 (1.0–3.0)	1.5 (1.0–2.3)	0.020 (0.000–0.054)	10

Table 16: Participant-level sensitivity on the confirmatory baseline, which holds all nine audited routes at ten races per risk level and so at the same matched size of 20. Columns, statistics, rarefaction and the 95% resampling interval (RI), which is not a confidence interval, are exactly those of Table 15; only the source of the trajectories differs, and the main paper’s admission verdict is printed beside each route. This table is the participant-level sensitivity for the primary dyad-clustered $q=0$ comparison. Every admitted route sits entirely below the human Hill $q=1$ interval at all three risk levels. The two routes that do not, GPT-5.4 mini and GPT-5.4 nano, are both routes the gate refused.

Risk	Population	n_{src}	Hill $q=0$ (95% RI)	Hill $q=1$ (95% RI)	Hamming (95% RI)	$n_{cluster}$
0.1	Human	98	19.2 (15.0–20.0)	18.9 (13.2–20.0)	0.467 (0.407–0.504)	98
	Gemini 3 Flash (admitted)	20	3.0 (1.0–3.0)	1.5 (1.0–2.3)	0.020 (0.000–0.054)	10
	Claude Opus 5 (admitted)	20	1.0 (1.0–1.0)	1.0 (1.0–1.0)	0.000 (0.000–0.000)	10
	GPT-5.4 (admitted)	20	5.0 (3.0–5.0)	4.0 (1.9–4.9)	0.171 (0.076–0.234)	10
	GPT-5.5 (admitted)	20	11.0 (3.0–11.0)	6.3 (2.2–8.9)	0.145 (0.058–0.216)	10
	Claude Sonnet 5 (admitted)	20	2.0 (2.0–2.0)	2.0 (2.0–2.0)	0.211 (0.211–0.211)	10
	Gemini 3.1 Flash Lite (refused)	20	3.0 (1.0–3.0)	1.5 (1.0–2.3)	0.020 (0.000–0.054)	10
	GPT-5.4 mini (refused)	20	20.0 (10.0–16.0)	20.0 (8.3–15.2)	0.416 (0.330–0.458)	10
	Gemini 3.5 Flash Lite (refused)	20	10.0 (6.0–10.0)	10.0 (5.0–9.5)	0.282 (0.200–0.331)	10
0.6	Human	104	19.7 (16.0–20.0)	19.7 (14.4–20.0)	0.495 (0.456–0.514)	104
	Gemini 3 Flash (admitted)	20	14.0 (6.0–13.0)	12.3 (4.9–11.7)	0.333 (0.251–0.363)	10
	Claude Opus 5 (admitted)	20	1.0 (1.0–1.0)	1.0 (1.0–1.0)	0.000 (0.000–0.000)	10
	GPT-5.4 (admitted)	20	11.0 (5.0–11.0)	8.4 (3.4–10.0)	0.257 (0.145–0.297)	10
	GPT-5.5 (admitted)	20	13.0 (5.0–13.0)	10.7 (4.2–11.5)	0.326 (0.259–0.352)	10
	Claude Sonnet 5 (admitted)	20	5.0 (4.0–5.0)	4.9 (3.0–5.0)	0.180 (0.132–0.205)	10
	Gemini 3.1 Flash Lite (refused)	20	7.0 (3.0–7.0)	5.8 (3.0–6.8)	0.186 (0.094–0.254)	10
	GPT-5.4 mini (refused)	20	17.0 (8.0–15.0)	16.2 (6.6–14.1)	0.360 (0.251–0.426)	10
	Gemini 3.5 Flash Lite (refused)	20	13.0 (7.0–13.0)	11.5 (4.8–11.7)	0.380 (0.290–0.411)	10
0.9	Human	138	19.5 (16.0–20.0)	19.3 (14.8–20.0)	0.497 (0.461–0.516)	138
	Gemini 3 Flash (admitted)	20	12.0 (5.0–12.0)	10.0 (4.0–10.7)	0.414 (0.349–0.419)	10
	Claude Opus 5 (admitted)	20	1.0 (1.0–1.0)	1.0 (1.0–1.0)	0.000 (0.000–0.000)	10
	GPT-5.4 (admitted)	20	14.0 (6.0–13.0)	12.5 (5.0–12.3)	0.414 (0.251–0.480)	10
	GPT-5.5 (admitted)	20	13.0 (5.0–12.0)	10.7 (3.9–10.7)	0.269 (0.155–0.349)	10
	Claude Sonnet 5 (admitted)	20	7.0 (5.0–7.0)	6.0 (3.2–6.8)	0.267 (0.152–0.331)	10
	Gemini 3.1 Flash Lite (refused)	20	7.0 (5.0–7.0)	6.4 (3.4–6.8)	0.305 (0.194–0.371)	10
	GPT-5.4 mini (refused)	20	17.0 (7.0–15.0)	15.2 (5.5–14.1)	0.408 (0.258–0.477)	10
	Gemini 3.5 Flash Lite (refused)	20	13.0 (7.0–13.0)	12.3 (5.6–11.7)	0.403 (0.316–0.417)	10
	GPT-5.4 nano (refused)	20	16.0 (8.0–14.0)	15.2 (6.3–13.2)	0.412 (0.267–0.465)	10

S7 Exploratory extensions

S7.1 Matched group-size comparison

Older three- to five-player pilots used a separately built lane and had no two-player arm, so they remain excluded from this evidence package. This subsection instead reports a matched comparison in which every group size, including two, runs through the same N -player engine, prompt template, parser and protocol. The number of seats is the only field deliberately varied.

With k of N companies choosing SAFE in a round, the engine assigns stage payoffs using the group-count rule of Han et al. [3]:

$$\pi_S(k) = \begin{cases} -c + \frac{b}{N}, & k = N, \\ -c + \frac{b}{k + s(N - k)}, & 1 \leq k < N, \end{cases} \quad (1)$$

$$\pi_U(k) = \frac{sb}{k + s(N - k)}, \quad 0 \leq k < N.$$

Here $b = 4$, $c = 1$ and $s = 1.5$; the source rule’s discovery term is set to zero. At $N = 2$ this returns the headline game’s four stage payoffs exactly: 1.0, 0.6, 2.4 and 2.0. SAFE and UNSAFE add 1.0 and 1.5

progress steps, respectively. The hidden horizon lasts at least five rounds and then stops after each completed round with probability 0.2. The prize $B = 100$ is split among the companies tied for greatest progress, and the same private-risk rule and three risk levels apply. The released design check asserts the $N = 2$ reduction and every invariant mechanism field before a task is collected.

What was collected, and how. The full grid: every group size at every private-risk condition, $p_r^{\max} \in \{0.1, 0.6, 0.9\}$, ten repetitions each, on the admitted route `google/gemini-3-flash-preview`. That is 120 races and 3,696 decisions with zero parse failures. No collection account available to us sustains the whole four-by-three grid, which needs roughly 7,800 requests, so the sweep was partitioned into the analysis unit itself, one (group size, risk) cell of ten repetitions, and each cell was collected whole on one identity so that it carries its own race-clustered interval rather than being stitched together across accounts. Cells were assigned to identities in a plan committed before the first was started, and the assignment rotates across risk levels so that no identity is confounded with a group size; every cell records its collecting identity in its collection receipt. Every cell’s bootstrap generator is derived from a digest of that cell alone, so collecting one condition cannot move an interval already reported for another, and the released analysis code refuses to report any risk level that is missing an arm.

Within each risk level, the four group-size cells were collected on four different accounts; the assignment rotates across risk levels so that every account collected each group size once. Account is therefore orthogonal to group size over the full grid, but a contrast within one risk level still combines group size with account-specific execution variance. The repeated contrast sizes across independently assigned risk levels reduce but do not remove that limitation. Each cell remains whole on one account and carries its own race-clustered interval.

The contrast is paired on the horizon. The game seed is base + rep and does not depend on the treatment name or on the number of seats, so one repetition index reuses a single horizon stopping-draw stream at every group size. Differencing a group size against the two-player arm inside a repetition therefore removes the horizon draw from the comparison. The analyser re-derives that pairing from the recorded game seed instead of assuming it, and it holds for all ten blocks in every contrast.

What it shows. UNSAFE play rises monotonically with the number of competing companies at every private-risk condition that leaves room for it to rise, and all six of those contrasts exclude zero. Because the two-player arm is the headline game itself, this also says something about the rest of the paper: the two-player results are the low end of that curve, not a special case beside it.

The table is deliberately kept separate from the presentation and seat checks. Group size is one route-specific exploratory manipulation, not a common scale on which the other robustness effects should be compared. Adding companies does not carry play up a plane. It carries play into the ceiling, and the five boundary cells are where the measurement stops being a measurement: those cells contained no safe decision at all, so their contrast against the two-player arm is bounded by the boundary and not by the endpoint.

Table 17: Matched group-size comparison on google/gemini-3-flash-preview. Ten races per cell, zero parse failures throughout. Rates carry 95% percentile bootstrap intervals over races; the contrast is the mean difference against the two-player arm within a repetition block, with its interval taken over the ten blocks. Horizon pairing was verified for 10 of 10 blocks in every row. Rates pool decisions within a cell. A contrast is the mean over matched repetitions of the within-repetition difference in race-level rates, so a contrast need not equal the difference of the printed rates. At $p_r^{\max} = 0.1$ every cell is at the ceiling, so its intervals have zero width: that is an absence of variation in the sample, not precision. The five-company cell at $p_r^{\max} = 0.6$ is at the ceiling too, so its contrast there is truncated, which leaves the high-risk condition as the only one that measures the five-company effect against no boundary.

N	Decisions	UNSAFE rate (95% CI)	minus N=2 (95% CI)	
<i>Low private risk, $p_r^{\max} = 0.1$</i>				
2	176	100.0% (100.0–100.0)	baseline	
3	264	100.0% (100.0–100.0)	+0.0 pp	(+0.0 to +0.0)
4	352	100.0% (100.0–100.0)	+0.0 pp	(+0.0 to +0.0)
5	440	100.0% (100.0–100.0)	+0.0 pp	(+0.0 to +0.0)
<i>Medium private risk, $p_r^{\max} = 0.6$</i>				
2	176	68.8% (65.2–72.4)	baseline	
3	264	80.3% (76.0–83.8)	+11.1 pp	(+6.3 to +15.1)
4	352	98.3% (95.6–100.0)	+28.2 pp	(+22.6 to +33.2)
5	440	100.0% (100.0–100.0)	+31.0 pp	(+26.4 to +35.3)
<i>High private risk, $p_r^{\max} = 0.9$</i>				
2	176	58.0% (50.7–63.7)	baseline	
3	264	68.9% (62.3–74.7)	+11.0 pp	(+5.9 to +16.4)
4	352	83.8% (80.6–86.2)	+28.1 pp	(+24.5 to +32.0)
5	440	96.1% (92.3–98.6)	+40.0 pp	(+33.2 to +47.4)

The size of the contrast is what makes the grid worth having. Going from two companies to three adds 11.1 points at $p_r^{\max} = 0.6$ and 11.0 at 0.9; going to four adds 28.2 and 28.1. The two risk conditions were collected on different days and different identities, and they agree to a tenth of a point. That agreement holds for the paired contrast. The differences of the printed rates at four companies are 29.5 and 25.9 points. The five-company contrast is the exception, +31.0 against +40.0, and the reason is visible in the table rather than interesting: at $p_r^{\max} = 0.6$ the five-company cell is already at 100%, so its contrast is truncated by the boundary and understates the effect. The high-risk condition is the only one that measures the five-company step against no ceiling.

The low risk condition is where the comparison becomes informative about its own limits. There every cell is at 100%, so the group-size contrast is exactly zero at every size. The right reading is not that group size stops mattering when private risk is low. It is that this route is already at the ceiling before any competitor is added, so the design has no room left in which the contrast could appear. The same route reaches 98.9% and 100.0% UNSAFE at $p_r^{\max} = 0.1$ in two independent neutral-baseline runs collected under a different protocol on a different engine, so the ceiling is a property of the

route at low risk rather than an artifact of this sweep. Read together with the ceiling at $p_r^{\max} = 0.6$ for five companies, the pattern is that this measurement has power only where play is off the boundary, which is a statement about the instrument and not about where the mechanism operates.

What it does not show. Group size cannot be separated from everything that comes with it, and no analysis repairs that. The group-count rule makes the stage payoff depend on how many companies there are, and the state description names each of them, so a larger group is also a different payoff and a longer prompt. The estimand is therefore a descriptive group-size contrast *within this game’s own definition*. The comparison covers one route and ten races per cell, so it is not a claim about endpoints or about a population of models. It is regenerated by the released analysis code, which refuses to report a risk level unless all four group sizes are present, refuses a cell with a parse failure or the wrong seat count, and records the collecting identity of every cell it uses.

AI-assisted technology statement

Claude Code and OpenAI Codex were used for code assistance, manuscript editing and verification. Exact model versions and complete interaction histories were not retained. The authors checked the resulting code and prose against the archived evidence and take responsibility for the manuscript, analyses, citations and released artifacts.

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